

Characterizing spatial functional microniches with SpaceTravLR

Koushul Ramjattun^{*1,2,3,5}, Alyson Wang^{*2,4,5,6}, Shilpi Giri^{#2,5}, Andrew Baessler^{#5}, Charlotte J. Imianowski^{#5}, Yijia Chen^{2,5,7}, Kennedy K Takehara⁸, Hannah Lee⁹, Jian Cui⁵, Yuchen C Liu^{2,4,5}, Chien-Sin Chen¹⁰, William A MacDonald¹¹, Jane C Siwek^{2,4,5,6}, Nathan D Lord⁶, Daniel H Kaplan^{5,10}, Maximilian Heeg⁸, Creg J Workman⁵, Dario AA Vignali^{5,12}, Amanda C Poholek^{^2,5}, Youjin Lee^{^1,2,3,5}, Jishnu Das^{^1,2,4,5,6}

¹Department of Biomedical Informatics, University of Pittsburgh School of Medicine, Pittsburgh, PA, USA

²Center for Systems Immunology, Department of Immunology, University of Pittsburgh School of Medicine, Pittsburgh, PA, USA

³Tumor Microenvironment Center, UPMC Hillman Cancer Center, Pittsburgh, PA, USA

⁴Joint CMU-Pitt PhD Program in Computational Biology, Pittsburgh, PA, USA

⁵Department of Immunology, University of Pittsburgh School of Medicine, Pittsburgh, PA, USA

⁶Department of Computational and Systems Biology, University of Pittsburgh School of Medicine, Pittsburgh, PA, USA

⁷Integrative Systems Biology PhD Program, University of Pittsburgh School of Medicine, Pittsburgh, PA, USA

⁸Allen Institute for Immunology, Seattle, WA, USA

⁹Survey Genomics, Inc, Berkeley, CA, USA

¹⁰Department of Dermatology, University of Pittsburgh School of Medicine, Pittsburgh, PA, USA

¹¹Department of Pediatrics UPMC Children's Hospital of Pittsburgh, Pittsburgh, PA, USA

¹²Cancer Immunology and Immunotherapy Program, UPMC Hillman Cancer Center, Pittsburgh, PA, USA

^{*}=co-first author/equal contribution

[#]=co-second author

[^]Corresponding authors – Jishnu Das (jishnu@pitt.edu), Youjin Lee (youjin.lee@pitt.edu), Amanda Poholek (poholeka@pitt.edu)

Abstract

The advent of spatial omics has revolutionized our understanding of tissue biology; however, these technologies remain largely descriptive and do not capture how changes in gene regulation propagate across spatial neighborhoods. While *in-silico* perturbation methods and foundation models aim to model the impact of genetic perturbations, these methods are limited to single-cell approaches that lack spatial resolution. Other studies can delineate morphological domains based on transcriptional similarity, but not spatial functional microniches. We address this major unmet need by developing SpaceTravLR (Spatially perturbing Transcription factors, Ligands and Receptors), a novel interpretable machine learning approach that generalizes across tissues and species, uncovering spatial features linked to functional outcomes, thereby capturing functional microniches with spatial resolution. SpaceTravLR infers how single or combinatorial genetic perturbations rewire signals across the tissue neighborhood, by propagating effects through underlying spatially resolved molecular networks, thereby modeling how perturbations can reshape both the targeted cell and its surrounding neighborhood. SpaceTravLR defines novel spatial microniches across a range of tissues at different scales of organization (niches, neighborhoods and tissues), disease and developmental contexts. SpaceTravLR's perturbation predictions are made solely from spatial omics data and closely align with experimental validation or known outcomes based on mechanistic studies. Critically, our approach enables the generation of mechanistic hypotheses underlying identified niches. We show SpaceTravLR discovered a novel transcription factor *Patz1* that synergizes with known inhibitory receptors (including *Pdcd1* and *Lag3*) to modulate T cell exhaustion in melanoma. SpaceTravLR also uncovered a mechanism for *Ccr4* that drives the spatial location of a pathogenic population of allergen-specific T helper 2 (Th2) cells as they develop in the lymph node, which was experimentally validated in a murine model. Overall, SpaceTravLR provides a novel interpretable and experimentally validated framework for uncovering how genes act individually and combinatorially through cell-intrinsic and cell-extrinsic circuits to shape spatial tissue organization and function. SpaceTravLR is available as open-source cross-platform standalone software, accompanied by extensive documentation.

Main

Cells can extend their transcriptome beyond their own cytoplasm via a network of secreted molecules and functionally impact the fate of neighboring cells¹ via surface receptors. A holistic understanding of tissue biology and organization therefore requires connecting each cell's molecular activity to its spatial context within the tissue². Combined with genomic perturbation tools such as CRISPR³, these advances hold enormous promise towards identifying causal modulators driving functional phenotypes, particularly when paired with high throughput single-cell RNA sequencing (scRNA-seq). Emerging technologies such as Perturb-seq⁴ and CROP-seq⁵ have already shown great potential for causal gene modulatory network reconstruction. However, these methods fail to recover the spatial context within which a perturbation occurs. To gain a holistic understanding of tissue biology and organization, we must link the spatial context of a cell to its biology using tools that capture interacting transcriptomes and perturb genes to identify causal drivers of function. Over the past year, a few cutting-edge platforms have started emerging that can successfully capture genetic perturbations in their intact spatial context. Perturb-FISH⁶ pioneered imaging-based spatial perturbation screening by coupling MERFISH⁷ with in situ amplification of guide RNAs. Perturb-Multimodal extended this concept in vivo by using MERFISH imaging to identify sgRNAs, mRNAs, and protein on the same tissue but with limited gene coverage. Spatial Perturb-seq⁸ delivered multiplexed in vivo CRISPR knockouts within single cells of intact tissue and is compatible with both sequencing- and probe-based spatial modalities, allowing whole transcriptome measurements. More recently, PerturbSpace⁹ and SPAC-seq¹⁰ showcased high-throughput sequencing frameworks that captures sgRNAs alongside spatial transcriptomes on platforms such as XYZeqv2¹¹ and Visium HD¹²/Stereo-seq¹³ respectively. As of now, these platforms are not commercially available and expensive to setup for exploratory investigations. Here, we introduce SpaceTravLR, a first-in-class framework to advance spatial biology at single cell resolution from descriptive mapping to causal modeling by uncovering functional microniches directly from spatial transcriptomics (ST) data. SpaceTravLR connects the downstream effect of gene perturbation to the spatial neighborhoods in which those effects occur, thereby linking molecular function to local tissue organization. The integration of *in-silico* perturbation screening with spatial context allows us to infer how the combination of gene-gene interactions rewire signaling to define distinct cellular environments.

We start by verifying the capabilities of SpaceTravLR by applying it to well-studied systems and show that we can accurately recapitulate spatially heterogeneous perturbation outcomes across T and B cell populations in the germinal center. We then highlight SpaceTravLR's unique ability to infer functional cell-cell communication using a new true single cell spatial murine kidney dataset, where we identify microniches associated with distinct macrophage predicted transition outcomes based on expression

of the ligand *Mif* in surrounding epithelial cells. We validate our model in the context of embryo development using a spatially profiled knockout dataset, showing accurate predictions for cell states unseen during training. To screen for novel gene candidates that can drive desired functional outcomes, we systematically perturbed all transcription factors, ligands and receptors across all cell types in human melanoma and a murine model of allergic asthma. In melanoma, we identify and experimentally validate a novel target that enhances CD8 T cell anti-tumor activity. In asthma we identify and experimentally validate an unconventional migration of allergen-specific pathogenic T helper 2 (Th2) cells in the lung draining lymph node following disrupted *Ccr4*.

An interpretable machine learning model for discovering functional microniches from spatial transcriptomics data

SpaceTravLR aims to push spatial analyses at single cell resolution from descriptive correlations toward functional mechanistic insights. To this end, we leverage spatial transcriptomics data to uncover functional microniches - tissue regions where local differences in the cellular microenvironment drive divergent cell fates or states.

To accurately model these complex dynamics, it is essential to capture both the intracellular regulatory programs that determine cell state and the intercellular signaling that organizes neighboring cells. To achieve this, SpaceTravLR learns from spatial transcriptomics data to directly link gene function and their perturbations to spatial phenotypes and signaling pathways. SpaceTravLR implements a scalable pipeline leveraging high-performance computing for parallelized tensor computations to estimate the target gene expression based on spatially varying regulatory and signaling dynamics. By leveraging convolutional neural networks¹⁴ (CNNs) to generate spatially resolved sparse graphs with differentiable edges (**Fig. 1A**), this architecture enables signals to propagate both within cells through regulatory edges and between cells through ligand-mediated connections, and is mathematically computed by efficient, gradient-based perturbation analysis via chain rule (**Fig. 1A, Supplementary Figs. 1-2**). Regulation is captured through TF terms, while signaling is modeled via distance-weighted ligand expression from neighboring sender cells to each receiver cell and incorporated into ligand-receptor (L-R) interactions and ligand-TF (L-TF) associations (**Fig. 1A, Supplementary Fig. 3, Methods**). While spatial variation is inherently resolved by the CNN, this formulation enables us to perform spatial propagation of cellular perturbations and trace their effects through complex regulatory and signaling networks. Before proceeding to comprehensive biological benchmarking, we first evaluated various alternative implementations of regularization and hyperparameters and also confirmed the model's baseline accuracy on simulated data, faithfully recovering the spatially

145 varying coefficients for the three modulator groups (**Supplementary Figs. 4-7**). Unlike
146 previous perturbation methods, SpaceTravLR can predict how genetic perturbations can
147 change a cell's own transcriptome as well as that of neighboring cells (**Figs. 1B, 1C,**
148 **Methods**).

149
150 We applied our model to 16+ datasets across diverse spatial transcriptomic technologies
151 and tissue types (**Extended Fig. 1**), showcasing wide applicability across different tasks
152 such as (1) inferring functional cell-cell communication events, (2) *in-silico* modeling of
153 functional and spatial reprogramming following changes in TFs, ligands and receptors
154 expression, (3) identifying spatial domains and functional microniches and their defining
155 genes, (4) utilizing transfer learning to identify functional microniches in novel tissues (**Fig.**
156 **1D**). We applied SpaceTravLR to different biological systems to evaluate whether it can
157 recapitulate known regulatory and signaling effects and uncover novel context-dependent
158 mechanisms in tissue. Our end-to-end computational pipeline and pretrained models
159 spanning multiple tissue types are made freely available as an open resource for the
160 scientific community (**Extended Fig. 2A**). Furthermore, we have ported SpaceTravLR
161 from Python to Rust, significantly enhancing computational efficiency, data preprocessing
162 and training to derivative computation and network propagation (**Extended Figs. 2B-C**).
163

164 **SpaceTravLR captures the impact of single and combinatorial** 165 **perturbations on spatially resolved cell-intrinsic and cell-extrinsic** 166 **circuits across tissue types**

167
168 We first sought to elucidate whether SpaceTravLR can recover established regulatory
169 and signaling effects using a spatially profiled single nuclei human tonsil dataset
170 generated using the Slide-tags¹⁵ platform. Tonsils have well-defined spatial architectures
171 and include complex mixtures of immune cells (CD8 T cells, different B and CD4 T cell
172 subsets, **Fig. 2A**) that work synergistically to mediate adaptive responses in a range of
173 contexts spanning natural and vaccine-mediated immunity¹⁶⁻¹⁸. Thus, characterizing
174 functional niches underlying these responses is of significant biological interest. We used
175 this dataset to assess our model's ability to predict how individual and combinatorial
176 perturbations influence spatially resolved molecular networks across tissue types. Then,
177 we extended this analysis to delineate spatial regions with distinct functional profiles,
178 defined by differences in perturbation impact.

179
180 SpaceTravLR captured the spatially varying and cell-type dependent effects on cell-
181 intrinsic and cell-extrinsic circuits across all 18 complex subclusters in this tissue context.
182 We quantitatively evaluated how to perturb a range of molecules across different cell
183 types. The selected molecules included lineage-defining TFs and key ligands and
184 receptors that are known to shape their cellular identities. We demonstrate

SpaceTravLR's ability to accurately capture the cell-type-specific responses to relevant TFs, ligands and receptor perturbations with high sensitivity and specificity, demonstrating our model's ability to resolve both cell-intrinsic and cell-extrinsic signaling-driven perturbation effects (**Supplementary Fig. 8**).

SpaceTravLR uniquely captures how localized perturbations influence neighboring cells (**Fig. 2B**). Our *in-silico* simulations demonstrate that overexpressing *Cxcl13* exclusively within the follicular dendritic cell population selectively alters the transcriptome of neighboring cells expressing the corresponding receptor *Cxcr5*¹⁹. We confirmed the predictive accuracy across other cell-type-specific perturbations, including *Il21* overexpression (OE) restricted to T follicular helper (Tfh) cells (affecting *Il21r*^{high} cells) and *Il18* restricted to myeloid cells (affecting *Il18r*^{high} cells)^{20–25}. Furthermore, SpaceTravLR successfully captured indirect signaling dynamics. A *Bcl6* knockout (KO) confined entirely to Tfh cells, which impairs their *Il21* production²⁶, triggered significant transcriptomic shifts within nearby cells as a direct function of their *Il21r* expression. Critically, SpaceTravLR predicts that OE or KO of these factors selectively affects only certain immune cell populations and not others. These in-silico simulations agree very well with prior biological expectation^{27–31}.

Next, we systematically benchmarked SpaceTravLR against Tahoe-x1³², CellPLM³³, SpaceFlow³⁴, and CellOracle³⁵, four recent state-of-the-art methods for inferring gene-gene networks. We chose these approaches as representative examples of conceptual paradigms widely used for *in-silico* perturbations. Tahoe-x1 is a single-cell foundation model trained on single cell data from drug perturbation assays, CellPLM is another foundation model trained on spatially resolved sequencing and imaging data while SpaceFlow is a graph based neural network that captures spatial dependency of gene expression profiles. All three methods generate informative cell-level embeddings that can be leveraged using a multi-layer perceptron for inferring the change in transcriptomic profile following a given genetic perturbation (see Methods). Meanwhile, CellOracle, is a popular linear model leveraging gene regulatory networks, but is limited to predicting TF perturbations only. Importantly, none of these methods incorporate complex spatial organization and cellular interactions that SpaceTravLR can leverage, and cannot predict how perturbations in one cell may impact neighboring cells. Thus, we created an alternative deep learning model to act as a preliminary spatial baseline³⁶.

Benchmarking against existing methods restricted our comparative evaluation to a limited subset of SpaceTravLR's features, since only SpaceTravLR can predict cell-extrinsic perturbation effects. Nevertheless, we evaluate every method's ability to resolve the quantitative effects of single and combinatorial gene perturbations, within a complex regulatory and signaling network. We modeled the perturbation combination of *Il21*, *Bcl6*,

and *Prdm1* with opposing roles in Tfh differentiation^{20,21}. *Il21* is a cytokine that promotes the Tfh lineage, while *Prdm1* and *Bcl6* are transcriptional regulators. By testing both perturbations separately or simultaneously, we can rigorously assess our model's capacity to capture how activators or repressors work within shared regulatory and signaling circuits. We quantitatively assessed our predictions by aligning the inferred perturbations along diffusion pseudotime (**Supplementary Fig. 9**). First, consistent with previous studies, SpaceTravLR predicted the expected effects: *Il21* OE, *Bcl6* OE, and *Prdm1* KO each individually promote Tfh differentiation^{37,38}. While *Bcl6* OE inhibits the T helper 1 (Th1) and T helper 2 (Th2) fates, *Prdm1* KO promotes the Th1 cell fate due to its role in Th2 differentiation^{39–41} (**Fig. 2C, Extended Fig. 3A**). In contrast, *Il21* KO, *Bcl6* KO, and *Prdm1* OE each individually inhibited Tfh differentiation. When these three regulators were jointly perturbed in pairs, the model predicted their antagonistic or synergistic relationships (**Fig. 2D, Extended Fig. 3A**). Since *Bcl6* and *Il21* both activate the Tfh lineage, OE or KO of both resulted in even stronger effects compared to either perturbation alone (**Fig. 2C**). When the two regulators were perturbed in opposite directions, *Bcl6* exerted a dominant effect, which aligns with its function being at the transcriptional level. These predicted interactions mirror known signaling circuits where *Bcl6* acts downstream of *Il21*⁴². In combinatorial perturbations involving *Prdm1*, *Il21* signaling dominated the Tfh cell fate predictions. However, because the Tfh and Th1 lineages are mutually exclusive, the overarching combinatorial predictions ultimately mirrored the expected Th1 differentiation trajectory⁴³ (**Extended Fig. 3A**). CD4 T cell differentiation is driven by branching developmental pathways and governed by complex, multi-layered regulatory networks, which our framework resolves by mapping regulatory relationships to predict cell fate outcomes following perturbation. While the alternative methods capture many of the individual and combinatorial results to some degree, SpaceTravLR performed significantly better than all of these approaches by more consistently recapitulating the expected results (**Extended Figs. 3B, 3C**). However, as with many computational models, SpaceTravLR's predictive accuracy decreases for extreme out-of-distribution values. When simulated OE exceeds two-fold the highest observed expression levels, the model yields aberrant downstream log2 fold changes and struggles to resolve distance-dependency with germinal centers (GCs) (**Supplementary Fig. 10**). Therefore, accurate simulation requires restricting perturbations to values proximal to the original data distribution⁴⁴.

Next, we moved beyond the impact of individual and combinatorial perturbations to examine how the spatial context influences cell fate, a critical step in defining spatial functional microniches in tissues. Tfh cells need continual extrinsic signals from GC B cells mediated via Cd40/Cd40l interactions^{38,45}. We examined the spatially-resolved combinatorial impact of three molecules promoting the Tfh lineage – *Stat3* (a TF), *Cxcr5* (a receptor) and *Il21* (a ligand). Interestingly, when we modeled *Il21* OE in Naive CD4 T

cells (Th0), the predicted *Stat3* and *Cxcr5* expression changes were highly dependent on the local GC B cell microenvironment (**Fig. 2E**). This relationship was non-linear, with correlation peaking at a threshold of 40 GC B cells, characterizing a spatial saturation effect where the signaling influence of the microenvironment reaches a limit despite increasing GC B cell abundance (**Supplementary Fig. 11**). This demonstrates the existence of functional microniches where *Stat3*, *Il21*, and *Cxcr5* synergistically promote Tfh fate only when sufficient GC B cells are present in proximity. Similar spatial dependencies were also observed for *Stat3*, *Il21*, and Tfh lineage promoting transcription factor, *Bcl6*²⁸ (**Fig. 2F**). Although CellPLM incorporates spatial information into its embeddings, it fails to resolve this critical biological dependency, further underscoring SpaceTravLR's unique capacity to explicitly parameterize and model localized cell-cell signaling dynamics. Thus, SpaceTravLR represents the first computational framework capable of simulating individual and combinatorial perturbations to resolve spatially dependent regulatory and signaling effects

Dissecting cell fate decisions dictated by the local environment

While existing *in-silico* methods can predict the differential impact of genetic perturbations across cell types/states, these methods are unable to uncover alternate cell fates for the same cell type/state based on spatial location⁴⁶. We used SpaceTravLR to infer the impact of genetic perturbations in modulating cell states across regions of varying spatial organization in complex tissues. This helps define functional microniches as changing spatial environments that can reshape cell type and behavior.

We focused on GC B cells, which cycle between dark zone (DZ) and light zone (LZ) through a gray zone (GZ) as they undergo affinity maturation within secondary lymphoid tissues^{18,31,47}. In the DZ, B cells proliferate and undergo somatic hypermutation (SHM), to increase antibody diversity and affinity, then migrate to the LZ to capture antigens to present to Tfh cells. High-affinity B cells receive survival signals and differentiate, while low-affinity clones return to the DZ for additional rounds of mutation and selection⁴⁸. A key TF driving this process in both GC B cells and Tfh cells is *Foxo1* (**Fig. 3A**). Building upon SpaceTravLR's ability to de novo identify spatially distinct regions within the GC (**Supplementary Fig. 12**), we evaluated whether the model could predict the differential spatial outcomes following *Foxo1* perturbation within the established DZ, LZ, and GZ compartments. SpaceTravLR predicted that *Foxo1* KO caused the DZ GC B cells to take on a LZ identity, but LZ GC B cells retained their identity despite exhibiting a similar transcriptional profile to DZ GC B cells (**Figs. 3B, 3C, Extended Fig. 3B**). This result aligns closely with the functional role of *Foxo1* characterized in murine models which demonstrate experimentally that *Foxo1* is critical for the spatial polarization and structure

of the GC, enabling GC B cells to re-enter the DZ, and that its loss diverts the cells to the LZ^{49–51}.

Critically, the predicted effects of *Foxo1* KO on tissue architecture were both interpretable and biologically meaningful. We first examined the primary L-R axes driving the expression of *Aicda*, which encodes the enzyme that governs somatic hypermutation within the dark zone, and is a defining molecular hallmark of DZ function. SpaceTravLR successfully recovered the expected, well-established pathways, including *Cxcr4*, *Cd40lg*, *Baff* (encoded by *Tnfrsf13b*) and *Il21* signaling (**Fig. 3D**). We then examined the other key genes and regulators that are known to control GC dynamics. In B cells, *Foxo1* positively regulates *Batf* and *Cxcr4*, which are essential for somatic hypermutation in the DZ through *Aicda*⁴⁹ (**Fig. 3A**). SpaceTravLR correctly simulated the increased expression of *Bcl6*, and decreased expression of *Batf* and *Cxcr4* in *Foxo1* KO (**Fig. 3E**), demonstrating gene-level precision and robust network propagation. SpaceTravLR achieves this level of mechanistic accuracy (established in murine models) with no explicit encoding of biological priors but solely based on interpretable model architecture which leverages the underlying biological network. These phenotypic and mechanistic changes identified by SpaceTravLR were not captured by CellOracle, a non-spatial, signaling-agnostic approach, but the best alternative method capable of making predictions without matched *Foxo1* knockout data (**Figs. 3F, 3G**).

SpaceTravLR learns the regulatory and signaling networks between cells in a specific spatial organization. We reasoned that these networks should be generalizable across different tissues and spatial profiling technologies, provided that the tissues contain the same cell types. We used our trained tonsil model to predict the effects of gene perturbations in the GCs of an unseen lymph node sample (**Fig. 3H**). While the tonsil was profiled with Slide-tags, the lymph node was profiled with Visium^{52,53}, a platform that captures a mixture of cells in one spot. Regardless, SpaceTravLR is still able to predict that *Foxo1* knockout in the lymph node causes a decrease in the spots with higher proportions of DZ cells, consistent with the shrinking of dark zones in experimental observations (**Figs. 3I, 3J**). The cross-context (across related biological tissues – tonsil and lymph node) and cross-platform generalizability of SpaceTravLR make it broadly applicable to capturing cell-intrinsic effects that lead to spatially resolved functional microniches.

Though *Foxo1* is well known for its role in B cells, its regulatory effects on Tfh cells are also well characterized. In mice, *Foxo1* knockout is known to remove repression of *Bcl6*, resulting in an expanded population of *Cxcr5*⁺ *Pdcd1*⁺ Tfh (bonafide GC Tfh) cells^{38,54,55}. Again, SpaceTravLR can make predictions based solely on the Slide-tags data, that upon *Foxo1* KO, the population of Tfh cells increases, as non-Tfh take on a Tfh transcriptional

identity (**Figs. 3K, 3L**). This occurs because of a concomitant increase in the expression of *Pdcd1*, *Cxcr5*, and *Bcl6* (**Fig. 3M**). In Th0, *Foxo1* has been linked to homing and survival of Th0 by regulating *Sell*, *Ccr7* and *Ilt7*⁵⁶. Simulated *Foxo1* knockout causes the downregulation of all three genes, consistent with experimental evidence (**Fig. 3M**).

These results utilized a CNN-based vision model which explicitly learns spatially dependent gene-gene relationships but is distinct from a transformer which may enhance learning spatially dependent relationships. Given the structural complexity of the GC, we sought to evaluate whether replacing our CNN-based vision model with a more sophisticated Vision Transformer (ViT) would enhance performance. Both architectures learned highly similar gene-gene relationships and achieved comparable performance in predicting target gene expression (**Supplementary Fig. 13**). Given their equivalent predictive accuracy, we selected the CNN-based model due to its substantially lower parameter count (i.e., less prone to overfitting even on sparse data increasing generalizability of our model across datasets) and reduced training costs. Thus, we find our CNN-based architecture effectively captures and incorporates spatial information at the level of other state-of-the-art vision models. To minimize spurious associations, SpaceTravLR first fits a sparse group lasso model to estimate global cell-type specific baselines (Methods). The vision model is then used to generate location-specific spatial multipliers, a factorization that isolates spatial signals from fundamental cell-type biology, improving model interpretability.

We further demonstrate this kind of factorization (isolating spatial signals from fundamental cell-type biology) in a murine model for tissue-resident memory (Trm) cell differentiation in the small intestine following LCMV infection. Virus-specific CD8 P14 T cells were adoptively transferred and the small intestine was profiled via a 350-gene Xenium panel at days 6, 8, 30 and 90 after infection⁵⁷ (**Extended Figs. 4A-B**). We used SpaceTravLR CNN to validate TGF β signaling, a known modulator of *Itgae* expression in P14 CD8 T cells⁵⁷ (**Extended Figs. 4C**). These predictions align closely with in vivo observations from a P14 CD8 T cell *Tgfb2* KO which exhibited an overall spatial distribution shifted toward the lower crypts alongside reduced *Itgae* expression. Subsequent examination of L-R interactions driving Trm activation markers recovered multiple canonical pathways, in particular with gene programs related to tissue migration, epithelial retention, and cytokine-mediated survival^{58–62}, while also uncovering a persistent *Ccl5-Ccr5* interaction across the four trained timepoints (**Extended Figs. 4D**). However, using an adoptive co-transfer model to study the effects of *Ccr5* of intestinal Trm differentiation revealed only minor phenotypic differences with a slight increase of immature CD103-CD69⁺ Trm cells at day 7 after infection. We speculated, that the false-positive prediction of *Ccr5* might be due to the limited gene panel in the Xenium experiment. Thus, we next applied SpaceTravLR to the same model system but profiled

with VisiumHD, which could measure a broader gene panel. Indeed, the *Ccl5-Ccr5* signaling axis was notably absent from the top L-R interactions (**Extended Figs. 4G-I**). Instead, it prioritized broader, more well-characterized activation pathways composed of genes not profiled in the Xenium panel. However, the spatial expression profiles of *Ccl5* and *Ccr5* within the VisiumHD architecture are highly correlated with the expression patterns of these top-ranked receptors, including integrins and granzymes (**Extended Figs. 4J-L**). Ultimately, while SpaceTravLR uncovered how *Ccl5-Ccr5* signaling influences Trm activation, the model only provides the most robust interaction prioritization when utilizing an expanded transcriptomic panel.

Next, we investigated SpaceTravLR's ability to move beyond genetic to chemical perturbations. We applied SpaceTravLR to a murine skin Xenium 5k panel profiling the initiation of cutaneous neurogenic inflammation, a process recently shown to unfold as a two-step, multicellular cascade in which Trpv1⁺ nociceptor-derived Substance P activates mast cells through MrgprB2, driving mast-cell-dependent dermal dendritic cell aggregation that is subsequently licensed by CGRP α to produce Type-17 cytokines⁶³. Because mast cells sit at the apex of this circuit, and because the dataset's pharmacologic perturbations directly manipulate this axis, capsaicin (Trpv1 agonism), compound 48/80 (a synthetic MrgprB2 agonist that activates mast cells⁶⁴), their combination, and untreated controls, it provides a stringent test of whether SpaceTravLR can resolve the signaling environment shaping mast-cell activation across conditions. Profiled on a Xenium Prime 5K panel, mast cells represent a small subpopulation, yet SpaceTravLR successfully recovered treatment-specific ligand-receptor programs promoting their activation and attributed each interaction to a defined cellular source, including DCs, fibroblasts, endothelial and epithelial cells, recapitulating the multicellular niche that the original study established through genetic dissection and imaging (**Extended Figs. 5A-E**). Across conditions, the model nominated the canonical mast-cell survival and activation axis Kitl-Kit⁶⁵ alongside extracellular-matrix-integrin/CD44 and complement-integrin interactions (e.g., C3-Itgam, C3-Itgb2, Icam1-Itgam), consistent with the dependence of mast cells on adhesive and stromal cues for their tissue positioning and reactivity⁶⁶. Importantly, rank-difference analyses distinguished the signaling landscapes of MrgprB2-driven (48/80, combo) versus Trpv1-driven (capsaicin) stimulation rather than producing a generic activation signature, mirroring the mechanistic specificity of the two-step model. Feature attribution by exact Shapley⁶⁷ decomposition showed that these predictions were jointly explained by ligand-receptor signaling (39.4%), transcription-factor activity (36.8%), and learned spatial context (22.3%), with negligible contribution from ligand-TF edges (1.4%), demonstrating that SpaceTravLR's inferences reflect genuine spatial neighborhood structure (**Extended Fig. 5F**). Together, these results show that SpaceTravLR is sensitive enough to resolve subtle, treatment-specific regulatory rewiring within a minor immune population in intact tissue, complementing the genetically

validated SP–MrgprB2–CCL2–CGRP α circuit with a transcriptome-wide, spatially resolved map of the interactions that assemble the neurogenic inflammatory niche. Notably, because SpaceTravLR models the signaling networks specific to the training sample, it serves as a highly sensitive platform for resolving minor regulatory shifts following mild pharmacological interventions. This capacity facilitates systematic, cross-condition comparative analyses to uncover differential signaling dynamics (**Extended Fig. 5G**).

Finally, we evaluated whether SpaceTravLR could identify tissue-level consequences at the phenotypic scale, which offers a more biologically relevant read-out than transcriptomic changes alone. We first applied the model to a Slide-seqV2 mouse embryo dataset⁶⁸, where *Tbx6*-deficient embryos classically develop ectopic neural tubes instead of somatic cells (**Extended Figs. 6A-B**). *Tbx6* is a transcription factor expressed in the presomitic mesoderm (PSM) and is required for somite partitioning and specification. To test whether our model could recover this developmental phenotype from ST data, we trained it on published WT embryo data and then performed *in-silico* *Tbx6* KO across all the cells at embryonic day 9.5 (E9.5), a well-studied stage critical to neural and brain development. Overlaying the ectopic tube transcriptomic profile, extracted from the *Tbx6* KO Slide-seqV2 dataset, onto the simulated perturbation transcriptomic profile, we observe this shift towards the ectopic phenotype (**Extended Figs. 6C-D**). Furthermore, across the top 1000 differentially expressed genes (DEGs) between the WT and KO, SpaceTravLR significantly outperformed in magnitude and direction of the expression changes compared to existing state-of-the art perturbation models and other benchmarks (**Extended Fig. 6E**). Our results demonstrate that SpaceTravLR can accurately infer the emergence of previously unseen cell states directly from ST data.

Because *Tbx6* and *Sox2* form an antagonistic feedback loop⁶⁹ that controls mesoderm to neural fate decisions, we simulated *Sox2* OE to test whether our model could infer this reciprocal developmental pathway. SpaceTravLR accurately predicted that *Sox2* OE drives somites to reacquire progenitor-like features (**Extended Fig. 6F**) and reproduced the expected shift in canonical neural tube patterning markers (**Supplementary Fig. 14**). Crucially, although the published KO data was generated at the zygote stage, our *in-silico* KO was performed at E9.5. Thus, while *Tbx6* KO more directly mirrors the observed phenotype, *Sox2* OE may be more faithful to the underlying developmental process, where the PSM fails to develop correctly and potentially becomes more NMP-like instead.

Disrupting functional paracrine signaling at single-cell resolution

While cell-intrinsic effects described above are a critical facet of signaling systems, they are often not sufficient to fully explain distal effects in tissue. SpaceTravLR also enables modeling of cell-extrinsic paracrine signaling networks, a critical and often underexplored facet of tissue organization. Existing *in-silico* methods for investigating cell-cell communication^{70–72} and microniche^{73,74} discovery often operate independently, overlooking a key biological principle that microniches shape cell function and behavior through intercellular signaling. To address this, we evaluated whether SpaceTravLR could capture paracrine signaling by isolating a perturbation in one cell type and quantifying its downstream impact on a spatially adjacent, unperturbed population. This is a complex problem of major biological significance not currently addressed by existing *in-silico* perturbation methods or foundation models.

For this, the kidney offers a great system for studying functional microniches and immune–microenvironment interdependence since the medulla contains varying regions of osmotic gradients. These hypertonic niches drive epithelial cells to produce chemokines that recruit resident and monocyte-derived phagocytes, and bias macrophages toward pro-inflammatory phenotypes under salt challenge^{75,76}. Here we set out to study how epithelial-immune cell interactions define and maintain different microniches. A functional understanding of cell-cell communication requires modeling the causal impact of ligand–receptor interactions on cell state at a single cell level. In collaboration with Survey Genomics, we developed a true single cell spatial profiling assay, XYZeqV2, based on a spatially hashed nanowell technology⁹. Briefly, by spatially barcoding a fresh slice of tissue prior to dissociation, we are able to recover the position of each cell using standard single cell workflows such as 10x. Thus, we can take advantage of the spatially resolved single cell data to probe cell-type specific gene expression and spatially contextualize receptor–ligand interactions. The functional relevance of the hypertonic macrophages identified here is their expression of inflammasome sensors such as *Nlrp1b*⁷⁷ which responds to high-salt conditions and subsequently triggers *Il1b* release during cellular stress⁷⁸ (**Extended Fig. 7A**). These hypertonic macrophages also act as salt sensors under homeostatic conditions, and alterations in their functions are directly related to delayed wound healing, and aberrant responses in infection and chronic disease such as atherosclerosis^{79,80}.

Macrophage migration inhibitory factor *Mif* is a key proinflammatory cytokine and chemokine-like function cytokine that is secreted predominantly from epithelial cells upon inflammatory and stress stimulation⁸¹ (**Fig. 4A**). Spatial mapping confirmed that these hypertonic macrophages are localized predominantly in the central kidney regions, where osmotic gradients are steepest, while regular macrophages occupy the periphery (**Fig. 4B**). These distinct populations appeared in other scRNA-seq murine kidney replicates,

but not other ST samples (**Supplementary Figs. 15-16**). As such, we predicted that the influence of *Mif* would be strongest within the central regions of the kidney, where the hypertonic zones were most apparent. In our murine kidney sample, *Mif* is primarily expressed by epithelial cells acting through several cell surface receptors expressed by other cell types, most notably myeloid cells, while *Cd74*, a high affinity receptor for *Mif*, is predominantly expressed by macrophages⁸² (**Fig. 4C**). Using SpaceTravLR, we modeled the impact of *Mif* KO in epithelial cells. SpaceTravLR successfully captures the signaling between *Mif* and *Cd74* and predicts substantial transcriptional changes in the macrophage population upon *Mif* KO only in the epithelial cells (**Fig. 4D**). Consistent with our hypothesis, the *in-silico* knockout caused the hypertonic macrophages to adopt transcriptomic profiles resembling regular macrophages. This reprogramming effect was absent in a Slide-tags kidney dataset lacking a hypertonic macrophage population, confirming the specificity of the transition (**Supplementary Fig. 16**). The *Mif*-mediated signaling network mediating this hypertonic-to-regular macrophage transition were undetected by NicheCompass's⁷³ enriched gene programs, underscoring the necessity of SpaceTravLR's direct signaling approach for resolving robust functional microniches (**Extended Fig. 7B**).

To evaluate SpaceTravLR's capacity for cross-sample transferability, we tested whether a model trained on a specific tissue specimen could generalize to independent samples of the same tissue type without retraining. We applied our trained CNNs from the first murine kidney dataset directly to a separate murine kidney dataset to generate the modulator coefficients for this second sample (**Fig. 4E**). The simulated effects of *Mif* KO generated by this model-transfer approach highly correlated with predictions from a SpaceTravLR model trained natively on that target tissue, demonstrating its capacity for zero-shot generalization of functional relationships across independent tissue samples (**Figs 4F, 4G**). Under both modeling paradigms, *in-silico* *Mif* KO in epithelial cells consistently induced a subpopulation of hypertonic macrophages to transition toward standard macrophage identities (**Fig. 4H**).

To convert the complex spatial signaling landscapes inferred by SpaceTravLR into high-level global microniches directly applicable to existing workflows, we sought to classify cells into discrete, non-overlapping functional domains (**Extended Fig. 7C**). We condensed the highly multi-dimensional regulatory coefficients into a single modulator activity matrix capturing the aggregate regulatory magnitude at each cell (Methods). Unsupervised clustering of this functional latent space partitioned the tissue architecture into discrete microniches, providing an automated framework to isolate and characterize cells sharing uniform spatial signaling profiles. We evaluated these functional microniches against benchmarking metrics established by NicheCompass (**Fig. 4I**), where both natively trained SpaceTravLR and our cross-sample model achieved best performance

overall, compared to BANKSY⁷⁴, COVET⁸³ (part of ENVI), and Nichecompass (**Extended Fig. 7D**). Remarkably, despite substantial variations in the global spatial organization between the two kidney samples, the transfer-learning model accurately recapitulated cohesive microniches that closely mirrored those identified by the natively trained target model (**Figs. 4J-L**). The global microniches identified by SpaceTravLR in the tonsil also outperform these state-of-the-art methods on the established benchmarking metrics, highlighting the robustness of a functionally driven approach to microniche identification (**Supplementary Fig. 17**)

SpaceTravLR accurately predicts how cell autonomous perturbations propagate to neighboring cells

By capturing how gene-level perturbations propagate through signaling networks, SpaceTravLR can model how cell autonomous perturbations alter their local environment. We quantitatively validate that our model accurately captures the downstream alterations across neighboring cells with experimental data as a gold standard validation. In a Spatial Perturb-Seq mouse hippocampus dataset, an in-vivo CRISPR technology⁸, multiple risk genes for neurodegenerative diseases were knocked out simultaneously within single cells of intact tissues using an AAV based method (**Fig. 5A**). To calculate the experimental DEGs, we compared the immediate neighbors of cells that received a specific targeted genetic perturbation (identified via the captured sgRNA) against control cells that did not receive any guide RNAs (**Fig. 5B**). To train SpaceTravLR, we used a publicly available mouse hippocampus Slide-seqV2 dataset and performed targeted in silico knockouts identical to the experimental guide targets⁸⁴ (**Fig. 5C**). As spatial Perturb-seq is an emerging technology with limited availability, the experimental yield is inherently sparse. Consequently, statistically significant DEGs could only be confidently extracted for specific spatial neighborhoods and target-neighbor cell-type pairings (**Extended Figs. 8A-C, Supplementary Fig. 18**). We compared the simulated significant DEGs in distinct hippocampal subfields (CA1/2/3) neighboring other CA1/2/3 cells that received *Clu*, *Fln*, *Fasn*, *Rrag*, *Srf*, and *Stk39* KOs, achieving high concordance with the experimental ground truth (**Fig. 5D**). We also report the correlation for all non-significant DEGs for rigor (**Extended Fig. 8D**). We additionally evaluated the DEGs in CA1/2/3 populations adjacent to CA1/2/3 targets subject to *Sh3gl2* and *Gfap* KOs, as well as the oligodendrocytes adjacent to oligodendrocytes that received *Stk39*, *Fln*, *Rrag*, *Trem2*, and *Olig2* KOs (**Supplementary Fig. 19A**). Again, the predicted and experimental transcriptomic changes exhibited high concordance.

We also compare SpaceTravLR's predictions against Perturb-Multimodal⁸⁵, an imaging-based technology for pooled genetic screening. Using the same pipeline described above,

we first compute the experimental DEGs of the cells in the neighborhood of the targeted cells in the murine MERFISH liver sample, following the whole transcriptome imputation described by the authors. We then trained SpaceTravLR on a homeostatic murine liver profiled with Stereoseqv2⁸⁶. While low expression levels restricted the robust evaluation of many transcripts, the model successfully resolved significant DEGs in hepatocytes surrounding *Bccip*-, *Lgr4*-, *Tbcb*-, and *Gpn1*-ablated targets. These predicted spatially constrained changes exhibited a strong positive correlation with corresponding experimental profiles (**Fig. 5G, Extended Figs. 8E**). An interesting observation from the Zhang et al.¹⁰ spatial CRISPR experiments on tumors is the highly conserved spatial localization dictated by functionally different microniches. The sgRNA-niche reported median Pearson's correlation was 0.6635¹⁰, with a mean cosine similar of 0.77¹⁰ across matched adjacent tissue slices. We sought to test whether the SpaceTravLR-inferred functional microniches from wild type observational data can accurately predict the perturbation-induced tumor microniches. By initially training SpaceTravLR on sections of tissue containing cells that had not received a gene perturbation, we identified global microniches that strongly align with those reported experimentally, which had conserved spatial localizations and associated gene programs (**Extended Fig. 9, Supplementary Fig. 20**). Following *in silico* knockouts mirroring the experimental guides that were enriched in at least two distinct niches, we found that the model's predicted TME enrichment scores were highly correlated with the experimentally observed log2 fold changes for *Cd83*, *Ptk6*, *Il4ra*, *Bcam*, *Cd74*, and *Cks1b* (**Extended Fig. 9**)

Despite significant progress in experimental spatial Perturb-seq, widespread adoption remains limited by data sparsity, expense, and platform accessibility. Given its robust capacity to accurately recapitulate these complex perturbation effects, SpaceTravLR provides a highly accessible computational framework for the broader research community. Ultimately, SpaceTravLR serves as a powerful *in silico* alternative to overcome the current translational bottlenecks of spatial perturbation screens.

SpaceTravLR enables rapid discovery and prioritization of genes that control cell fate in different microenvironments

Next, we sought to leverage SpaceTravLR's unique ability to capture the cell-extrinsic effects of targeted perturbations to systematically screen for genetic targets on a human melanoma dataset to engineer cellular states that alter immune-tumor interactions. Despite showing remarkable promise, checkpoint blockade immunotherapy is only durable for a small subset of patients^{87,88}. Several factors limit the success of T cell immunotherapy in solid cancers, including cell trafficking to the tumor, exhaustion due to chronic antigen exposure, and other immunosuppressive mechanisms operating within the tumor microenvironment (TME)^{89–91}. Here, we sought to identify key genes in CD8 T

cells, that restrain CD8 T cell exhaustion and promote anti-tumor phenotypes, by systematically perturbing all expressed TFs, ligands, and receptors. Using a Slide-tags human melanoma sample¹⁵ (from a patient prior to receiving immunotherapy) with a complex mixture of immune (monocytes, B, Treg, CD4 T, CD8 T) and tumor cells (**Figs. 6A, 6B**), we aimed to identify genes that could enhance CD8 T cell cytotoxicity while reducing tumorigenicity⁹². We mapped marker genes to spatial vector fields to define three key metrics – tumorigenicity, T cell exhaustion, and T cell activation (**Figs. 6C, Methods**). In the tumor compartment, vectors aligned along the tumorigenicity axis capture transitions from less aggressive to more aggressive and invasive phenotypes. In the CD8 T cell compartment, analogous gradients reflect activation-to-exhaustion dynamics, with vector directions representing the continuum from naïve or effector-like to terminally exhausted states (**Figs. 6D**). We superimposed these cellular fate outcome vectors with the SpaceTravLR simulated spatially resolved perturbation vectors to find gene targets in CD8 T cells that simultaneously rescue exhausted T cells, enhance T cell cytotoxic activity and decreases tumorigenicity. This demonstrates SpaceTravLR's utility as a screening tool for the identification of novel cell fate programs.

As expected, KO of known checkpoint inhibitory receptors (*Pdcd1* and *Ctla4*) enhanced anti-tumor activity^{93–96} (**Fig. 6E**), confirming SpaceTravLR's predictive accuracy. This is in line with their well-studied roles, with *Ctla4* and *Pdcd1* being primary targets of current checkpoint inhibitors currently widely used in the clinic. Our predictions of perturbations with more modest effects on tumor outcome are highly context-dependent, specifically *Npas3* KO and *Ccr1* OE on CD8 T cells, also yielded reductions in tumorigenicity scores and aligned with experimental observations (**Supplementary Fig. 21**). We then looked at more complex combinatorial effects of perturbing TFs and receptors. Consistent with previously published experimental results⁹⁷ in lymphoma, *Prdm1* KO reversed exhaustion, but failed to enhance T-cell cytotoxicity against tumors, likely due to the reported compensatory upregulation of *Nr4a3* (**Fig. 6F**). Furthermore, our simulation revealed that *Nr4a3* KO marginally increased anti-tumor activity^{98,99}. However, consistent with expectation, *Prdm1* and *Nr4a3* double KO synergistically prevented T-cell exhaustion while improving anti-tumor outcome (**Fig. 6F**). Beyond known regulators, our model discovered a putative novel target, *Patz1*, that upon knockout, is predicted to confer high anti tumorigenicity and restrain exhaustion (**Fig. 6G**). Intriguingly, the *Tyrp1* expression on the tumors, which is indicative of a highly proliferative and malignant population of melanoma cells¹⁰⁰ decreased proportionally with localized CD8 T cell number (**Fig. 6H**).

While *Pdcd1* and *Lag3* deletion and mAb blockade have been extensively studied due to their impact on T cell exhaustion and tumor outcomes¹⁰¹, the impact of *Patz1* knockout on intratumoral CD8 T cells remains mostly unexplored, although earlier studies suggest involvement in T cells lineages^{102,103}. Combination therapy involving relatlimab (rela; anti-LAG-3) and nivolumab (nivo; anti-PD-1) has been successful in improving outcomes in

metastatic melanoma¹⁰⁴. Prior work by us has uncovered corresponding mechanisms of combination relat and nivo therapy in patients from CD8 T cell-restricted deletion of *Pdcd1/Lag3* models¹⁰⁵. Using scRNAseq data of adoptively transferred CD8 pMEL cells into B16-gp100 tumor-inoculated mice from the latter, we show that wildtype mice possessed significantly upregulated *Patz1* expression relative to *Pdcd1* single- or *Pdcd1/Lag3* double deletion which exhibit a less exhausted phenotype¹⁰⁴ (**Figs. 6H-J**). Prompted by these findings showing that *Patz1* is a key transcription factor in T cell exhaustion along with SpaceTravLR's prediction for *Patz1* knockout, we performed a targeted knockout experiment using CRISPR-mediated gene editing in pMEL cells followed by adoptive transfer into mice prior to B16-gp100 inoculation. Multivariate analysis of intratumoral CD8 T cell analyzed by flow cytometry revealed that *Patz1* deletion indeed induces a phenotypic shift indicative of reduced exhaustion and heightened activation, consistent with SpaceTravLR inferences, with a linear SVM discriminator achieving 91.7% cross-validation accuracy ($p = 0.025$). (**Figs. 6K-M, Supplementary Fig. 22**).

Despite being able to extract highly actionable biological insights from a single tissue sample of just 5000 cells, SpaceTravLR has been engineered from the ground up to be scalable to millions of cells, thus providing a futureproof method for the growing scale of spatial datasets. To showcase this, we used two recently released human cancer datasets profiled using the Atera¹⁰⁶ platform from 10x genomics, allowing whole transcriptome measurements from hundreds of thousands of cells. SpaceTravLR processed the full Atera WTA cervical dataset¹⁰⁷ (**Extended Fig. 10A**) (717,576 segmented cells $\times$ 18,028 genes) without downsampling. Here our vision model captures the key micro-anatomically restricted signals around the peritumoral CAF belt abutting the p16⁺ core. At this resolution, we also infer SP1 as a dominant positive modulator of LUM specifically in CAFs immediately adjacent to the malignant core—consistent with LUM⁺ matrix CAF states that remodel ECM and promote invasion in cervical cancer—while IRF8 signaling was confined to differentiated tumor domains, supporting a squamous differentiation program rather than a stromal immune one (**Extended Fig. 10B**). Independently, a proliferation-focused ligand–receptor screen recovered canonical growth-promoting axes, including ECM–integrin, BMP7–BMP2, LTB–LTBR, and GAS6–AXL, with ligand sources enriched in fibroblasts and tumor-core populations—linking stromal ECM delivery and core-derived morphogen signaling to proliferative tumor programs (**Extended Fig. 10C**). The same architecture generalized to an independent FFPE breast cancer Atera WTA section (170,057 cells)¹⁰⁸ (**Extended Fig. 10D**). Here, region-specific modulators of CDH1 (E-cadherin) were enriched in ER⁺ tumor epithelium, consistent with maintenance of luminal adherens-junction programs in hormone-receptor-positive domains¹⁰⁹, whereas CCDC125-associated regulatory activity localized to tumor borders—suggesting a border-enriched epithelial program at the invasive interface rather than uniform expression across the tumor mass (**Extended Fig. 10E**). For MUC1⁺ niches,

top inferred interactions were dominated by epithelial adhesion and juxtacrine signaling—
LAMA3–CD44¹¹⁰ (basement-membrane anchoring), JAG1–NOTCH2 (epithelial
crosstalk), and CDH1–CDH1 homophilic adhesion—with ligand sources spanning ER⁺
tumor, epithelial, fibroblast, and border compartments. Together, these results
recapitulate how MUC1⁺ epithelial niches^{111,112} are structurally organized by laminin–
integrin anchoring¹¹³, Notch-mediated neighbor communication, and cadherin-mediated
cohesion in the breast tumor microenvironment (**Extended Fig. 10F**).

Our model permits unbiased virtual functional screens that are quick, scalable, accurate
and a tiny fraction of the cost of performing them experimentally to uncover actionable
perturbations that can potentially alter cell fate and transitions within the multicellular
spatial neighborhoods.

SpaceTravLR uncovers a novel Ccr4-dependent spatially restricted Th2 microniche that influences lung migration in allergic asthma

Finally, we sought to use SpaceTravLR to study spatial cell dynamics in a setting where
we could experimentally validate the model predictions. We focused on a murine model
of allergic asthma (**Fig. 7A**) to elucidate the functional roles of key molecules in
determining spatial migration patterns of T cell subsets. Allergic asthma is driven by a
population of pathogenic Th2 cells that migrate from the mediastinal lymph nodes
(medLNs) to the lung¹¹⁴. Specifically, our prior work¹¹⁵ demonstrates that Th2 cells require
signaling mediated by synergy between *Il10* and *Il2* and the corresponding receptors for
Il2 – *Il2ra*, *Il2rb*, and *Il2rg*. Here we used SpaceTravLR to dissect the underlying the spatial
migration of these pathogenic cells (**Figs. 7A, 7B**). To this end, we trained SpaceTravLR
on a Slide-seqV2 dataset with paired TCR collected from the medLN three days after
adoptive transfer of house dust mite (HDM)-specific TCR transgenic CD4 T cells followed
by intranasal administration of HDM daily for three days. These CD4 T cells were derived
from a T cell receptor (TCR) transgenic mouse specific to Derp-1 (1DER), a class II-
restricted immunodominant peptide from HDM¹¹⁶. Since 1DER T cells expand in vivo in
response to antigen challenge, they comprise a notable fraction of the Th2 population
within the lymph node. Spatial mapping in the medLN revealed anatomical organization
of T cell zones, B cell follicles, and TB borders, providing the necessary backdrop for
interpreting localizations of 1DER CD4 T cell subsets (**Fig. 7C**).

Using SpaceTravLR, we sought to identify the strongest spatially dependent ligand-
receptor interactions that controlled Th2 identity. Consistent with earlier findings¹¹⁵, we
recapitulated the importance of IL-2 axis in allergen-specific Th2 cell differentiation while
also identifying *Ccr4*-dependent ligand interactions involving *Ccl17*, *Ccl22*, and *Ccl5*<sup>117–
¹¹⁹ (**Fig. 7D**). Interestingly, the distribution of the *Ccr4*-*Ccl5* and *Ccr4*-*Ccl22* interactions</sup>

exhibited distinct spatial patterns, with Th2 cells closer to the central tissue region showing the highest scores (**Fig. 7E, Extended Figs. 11A**). Further, when the LR interactions for each zone were investigated separately, these interactions ranked most highly on the T cell zone, and less highly in the B cell follicles, mirroring the observed spatial gradient (**Extended Figs. 11B**).

To better elucidate the role of *Ccr4* in the context of functional microniches, we next performed *in-silico* deletion of *Ccr4* using SpaceTravLR and compared it to CRISPR/Cas9-mediated experimental knockout in 1DER cells (**Extended Fig. 11C**). This comparison establishes a very rigorous benchmark, as the perturbation is confined to a very specific cell population while preserving the broader lymph node structure. Although baseline cellular proportions vary due to modality and the spatial plane of the tissue slice, the global cell-type composition and zonal proportions (B-cell follicles, T-cell zones, and TB borders) are comparable across the four lymph node slices (one WT SlideseqV2, one WT VisiumHD, and two *Ccr4* KO replicates) (**Extended Figs. 11D-I**). Although the WT VisiumHD sample contained a higher fraction of B cells, we confirmed that all annotated cell populations robustly expressed their canonical lineage markers (**Extended Fig. 11J**). Finally, we also verified that *Ccl5* and *Ccl22* ligand expression remained heavily concentrated in the T-cell zones across all samples (**Extended Fig. 11K**).

Our model predicted that *Ccr4* deletion would cause antigen-specific Th2 cells in the T cell zone to become transcriptionally more similar to cells at the TB border and B cell follicles (**Figs. 7F, 7G**). While *Ccr4* is known to promote migration of Th2 cells to the lung¹²⁰, the impact of *Ccr4* deletion on its localization in the lymph node had not been described. The experimental knockout results directly validated the spatial redistribution of Th2 cells, which were significantly enriched in the B-cell follicles and depleted in the T-cell zones (**Fig. 7K**). This directional spatial shift is not driven by compositional bias, given that baseline B-cell frequencies were higher in the control cohort, nor is it driven by differences in the spatial localization of *Ccl5* and *Ccl22* expression (**Extended Figs. 11H-K**).

We next asked whether SpaceTravLR could capture how antigen-specific (1DER) *Ccr4*-deficient T cells influenced their local microenvironment. We evaluated the DEGs of the CD8 T cells, B cells, and DCs neighboring Th2 cells across control and *Ccr4* KO samples (**Fig. 7L**). While no existing computational method can fully disentangle these complex relationships, SpaceTravLR accurately recapitulated the transcriptomic shifts within the Th2 neighborhood, demonstrating its capacity to capture the nuanced, spatially dependent effects of *Ccr4* deletion in the local microenvironment (**Figs. 7M-O**). These findings suggest *Ccr4* deletion on HDM-specific CD4 T cells may disrupt centrally located Th2 cells and SpaceTravLR predictions can explain how local perturbations can affect regulation and signaling in microniches that impact actual biological functions.

Discussion

The ability to mechanistically connect biological function to spatial organization remains a major challenge in understanding multicellular systems. Recent developments in computational approaches can now generate predictions on cell-intrinsic perturbation effects, but they are not designed to capture how these effects propagate across cellular neighborhoods. Here, we introduced SpaceTravLR, an interpretable machine learning framework that addresses this gap by integrating spatially varying gene regulatory and signaling networks to explain how molecular perturbations shape both cell intrinsic and extrinsic circuits in tissue microenvironments. In doing so, SpaceTravLR moves spatial biology from a descriptive framework into one that can dissect causal relationships, enabling discovery of functional microniches, a localized cell cluster connected by causal regulation rather than spatial proximity or similar transcriptional states.

Moreover, through applications across diverse tissue contexts, species, and platforms, SpaceTravLR reveals that the impact of genetic perturbation is both spatially constrained and context dependent. Unlike other single-cell perturbation frameworks, our model learns spatially resolved gene-gene relationships and propagates the effects through both transcriptional and signaling networks. This design allows the model to infer *how* a perturbation within one cell type drives transcriptomics and spatial programming in its local tissue neighborhood. As a result, we can interpret these changes through known biological parameters to enable testable hypotheses that link molecular regulators to spatially resolved functional outcomes. Application of our model was able to reproduce known biology from *Foxo1*-mediated germinal center maintenance to *Tbx6* dependent embryo development. It extended to uncovering the novel TF target, *Patz1* that drives a phenotypic shift in CD8 T cells, and a previously unrecognized spatial reorganization and retention of *Ccr4*-dependent Th2 cells within the mediastinal lymph nodes of a murine model of allergic asthma. In each case, SpaceTravLR was able to broadly infer how intercellular communication shapes tissue architecture and cellular localization.

SpaceTravLR opens up multiple different avenues for expanding capabilities as spatial transcriptomics technologies continue to evolve. We anticipate that SpaceTravLR 2.0 will support training and inference of tissues profiled in their 3D context, allowing us to dissect ligand-receptor interactions beyond 2D and thus uncover 3D functional microniches. Current spatial platforms capture regulatory state, but many functional signals are executed through proteins, phospho-signaling, secreted ligands, chromatin accessibility and tissue morphology. Future versions of SpaceTravLR could integrate high-plex platforms such as GeoMx Digital Spatial Profiling, CosMx Spatial Molecular Imaging, CODEX and imaging mass cytometry to jointly model RNA, protein, histology and signaling activity. In the future, we also plan on developing a web interface to allow

researchers to leverage pretrained SpaceTravLR model to design perturbation experiments and to fine tune their own datasets cost effectively. We anticipate that SpaceTravLR can serve as a foundation for *in-silico* investigation that charts how gene pathways govern tissue organization and spatial microniches.

Methods

Nomenclature

- **Functional microniche:** a discrete tissue region whose cells share similar regulatory and signaling activity and therefore tend toward similar phenotypic outcomes when perturbed.
- **Global microniche:** continuous, non-overlapping clusters identified by using the SpaceTravLR's estimated coefficients of their overall regulatory and signaling network.
- **Transcriptomic shift:** the total change in all gene expressions in each cell after a simulated perturbation, relative to the unperturbed baseline.
- **Transition probability:** the likelihood that a cell adopts a new transcriptomic profile more correlated with a different cell state than its current one.
- **Perturbation:** an intrinsic (genetic) or extrinsic (chemokines/drugs) intervention that shifts the gene expression of one or more genes.
- **Regulator:** a transcription factor directly controlling the expression of its target genes.
- **Modulator:** non-transcription factor genes such as chemokines that can control the expression of downstream genes.
- **Propagation:** an iterative process by which an initial signal or perturbation spreads.
- **Cell-autonomous:** a process or event that originates and is constrained entirely within the cell, independent of external signals.

846

847 **Spatially informed signaling inference**

848 We start by building a continuous 2D ligand-receptor interaction vector field for each pair
849 using the coexpression of ligand and receptor genes within close spatial proximity. For
850 each receiver cell, we calculate the signaling activities of every ligand–receptor (L–R) pair
851 as the product of receptor expression and a distance weighted ligand expression^{121,122}.
852 The amount of ligand received on a cell surface is modeled as an aggregate of ligand
853 expression across the tissue, approximating molecular diffusion and enabling both
854 autocrine and paracrine constraints. Receptors are treated as membrane-anchored and
855 non-diffusible and are thus evaluated only on the receiving cell.

856 The effective signaling activity of pair k at cell i is

$$857 \text{LRSignal}_i^k = wL_i^k \times R_i^k, \quad (1)$$

858 where R_i^k denotes receptor expression and wL_i^k the spatially diffused ligand field at i .

859 The received ligand field is

$$860 wL_i^k = \frac{S}{|J|} \sum_{j \in J} L_j^k \times \Omega_k(i, j), \quad (2)$$

861 where S is a scaling hyperparameter (default=1), J is the number of cells involved, and
862 $\Omega_k(i, j) = \exp(-d_{ij}^2/(2\ell_k^2))$ is a Gaussian kernel with characteristic length ℓ_k . An optional
863 cutoff $d_{\max}$ sets $\Omega_k(i, j) = 0$ when $d_{ij} > d_{\max}$, restricting interactions to biologically
864 plausible ranges⁷². Longer ℓ_k is assigned to secreted ligands; shorter ℓ_k to contact-
865 dependent signaling, 250 and 30 microns by default respectively.

866

867

868 **Spatially informed regulatory network inference**

869 SpaceTravLR cannot determine if a given gene is a transcription factor, ligand, or receptor
870 using gene expression data alone. Therefore, we start by assembling a prior modulatory
871 network linking each target gene to candidate transcription factors (TFs), ligands, and
872 receptors. TF–target edges were downloaded from the published CellOracle database
873 while ligand–receptor and ligand–TF edges were obtained from CellChat and NicheNet¹²³
874 respectively. Next, we make this initial network more tissue and context specific by only
875 including gene-gene edges that are empirically supported by the available spatially
876 profiled single cell data. TF–target interactions are therefore screened using CellOracle
877 while ligand–receptor interactions are optionally screened with COMMOT⁷², by permuting
878 cell locations to obtain spatial null p -values (threshold $p < 0.3$ to retain a broad candidate

set of ligand-receptor interactors). CellOracle and COMMOT are used only for edge filtering; their coefficient estimates are not used by our model.

Target gene expressions in cell i are modeled as

$$\mathcal{H}_i = \beta_i^0 + \sum_{TF \in \mathcal{T}_{\mathcal{H}}} \beta_i^{TF} TF_i + \sum_{(L,R) \in \mathcal{P}_{\mathcal{H}}^{LR}} \beta_i^{L,R} wL_i^L R_i^R + \sum_{(L,TF) \in \mathcal{P}_{\mathcal{H}}^{LTF}} \beta_i^{L,TF} wL_i^L TF_i, \quad (3)$$

where β_i^0 is a location-specific intercept, β_i^X the coefficient for module X , and $\mathcal{T}_{\mathcal{H}}$, $\mathcal{P}_{\mathcal{H}}^{LR}$, and $\mathcal{P}_{\mathcal{H}}^{LTF}$ the TF, ligand–receptor, and ligand–TF candidate sets for target gene $\mathcal{H}$.

To obtain closed-form partial derivatives for perturbation propagation, each signaling term is restricted to two multiplicative factors. Heteromeric receptors are decomposed into constituent ligand–receptor pairs (e.g., $L \cdot R_a R_b \mapsto L \cdot R_a + L \cdot R_b$).

Spatially weighted regression (SWR)

The SWR¹²⁴ couples a custom sparse group lasso¹²⁵ with a convolutional neural network to estimate cell-specific regulatory coefficients while preserving interpretability and differentiability. This specific formulation then allows SpaceTravLR to propagate the perturbation signal through both cell intrinsic and cell extrinsic networks via the chain rule.

For each target gene and cell type c , we estimate mean regression coefficients by sparse group lasso:

$$\hat{\beta}_c^{\text{SGL}} = \underset{\beta}{\operatorname{argmin}} \frac{1}{2n_c} \|\mathbf{y}_c - \mathbf{X}_c \beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_G \sum_{\ell} \|\beta_{G_{\ell}}\|_2 \quad (4)$$

where $\mathbf{X}_c$ describes TF, received-ligand $\times$ receptor, and ligand–TF predictors; G_{ℓ} indexes regulator groups; the ℓ_1 penalty induces within-group sparsity. Compared to a vanilla group lasso regression where entire groups are pruned, the sparse group lasso allows independent terms to be zero-ed out within groups. Our implementation uses the FISTA algorithm with proximal ℓ_1/ℓ_2 updates for optimization.

$$R_c^2 = 1 - \frac{\sum_{i \in c} (y_i - \hat{y}_i)^2}{\sum_{i \in c} (y_i - \bar{y}_c)^2} \quad (5)$$

To prevent poorly performing models from polluting the gene network with bad coefficients we zero out cell type specific coefficients below an R^2 threshold (default=0.2). Optimally, SpaceTravLR supports adding additional families of genes by defining new groups to go beyond TFs, ligands and receptors. However, this setup can often lead to more spurious network edges.

909 Spatial context extraction

910 A spatial proximity grid (SPG) tessellates the tissue area into an $m \times n$ rectangular-
911 bounded lattice with pixel centres

$$912 \quad g_j^x = x_{\min} + (j + 1/2)\Delta_x, \quad g_i^y = y_{\max} - (i + 1/2)\Delta_y, \quad (6)$$

913 where $\Delta_x = (x_{\max} - x_{\min})/n$ and $\Delta_y = (y_{\max} - y_{\min})/m$. For cell j , the inverse-distance
914 map $\mathcal{D}_j$ assigns each pixel the value $1/\max(\| (x_j, y_j) - (g^x, g^y) \|_2, \epsilon)$ with $\epsilon = 10^{-6}$,
915 yielding a smooth radial encoding of tissue architecture at single cell resolution. For k
916 cell-type clusters, maps are stored as a $k \times m \times n$ tensor.

917

918 Vision encoder

919 To process the input SPG, we implement two backbones: a three-layer convolutional
920 network with batch normalization, parametric ReLU activations, and spatial pyramid
921 pooling; and a Vision Transformer that tokenizes the grid into contiguous patches and
922 applies multi-head self-attention.

923

924 Spatially varying coefficients

925 A multilayer perceptron maps the spatially encoded representation to location-specific
926 multipliers. Effective coefficients are computed as

$$927 \quad \beta_j = \beta^{\text{SGL}} \odot f_{\text{spatial}}(\mathcal{D}_j), \quad (7)$$

928 and expression is read out linearly,

$$929 \quad \hat{y}_i = \beta_{i,0} + \sum_{j>0} \beta_{i,j} x_{i,j}. \quad (8)$$

930 This factorization separates global regulatory structure from spatial modulation. The
931 network is trained end-to-end with an Adam optimizer to minimize the mean squared
932 error between the predicted and observed target gene expression.

933

934 Transferring learned coefficients from high to low resolutions

935 SpaceTravLR uses a vision model to learn regulatory and signaling networks within and
936 between cells, which generalize across tissues and spatial profiling technologies when
937 the same cell types are present, even if local spatial organization differs¹²⁶. For matched
938 resolution and modality, trained CNNs directly predict modulator coefficients; to transfer
939 from higher-resolution data (Slide-tags, XYSeqV2) to lower-resolution 10x Visium, we

align neighborhoods of five single-nuclei cells to each 55 μm spot (typically 1–10 cells). On the inference sample, Cell2location provides per-spot cell-type scores; each spot is matched to the training centroid with the most similar neighborhood structure (Pearson correlation), and cells receive the mean coefficients from that neighborhood. Once aligned, test samples support perturbation analyses in the same way as native-resolution data.

Spatial perturbations

SpaceTravLR simulates knockouts and overexpression by iterative propagation through the inferred GRN and the spatial LR signaling field.

The target gene to be perturbed is set to the final desired level, and a delta tensor is generated as follows:

$$\Delta_{i,g}^{(0)} = \begin{cases} y_i^{\text{des}} - y_i^{\text{base}} & g = g^* \\ 0 & \text{otherwise,} \end{cases} \quad (9)$$

where g^* is the perturbed gene.

At iteration t , partial derivatives of predicted expression with respect to modulators are evaluated from SWR coefficients and the current expression state. For target gene $\mathcal{H}$ at cell i ,

$$\frac{\partial \mathcal{H}_i}{\partial x_{TF}} = \beta_i^{TF}, \quad \frac{\partial \mathcal{H}_i}{\partial x_R} = \beta_i^{L,R} w L_i^L, \quad \frac{\partial \mathcal{H}_i}{\partial x_L} = \beta_i^{L,R} x_R \quad (10)$$

for ligand–receptor pairs (L, R) , and

$$\frac{\partial \mathcal{H}_i}{\partial x_{TF}} += \beta_i^{L,TF} w L_i^L, \quad \frac{\partial \mathcal{H}_i}{\partial x_L} += \beta_i^{L,TF} x_{TF} \quad (11)$$

for ligand–TF pairs, with contributions summed over edges on the same modulator.

We can then propagate changes to different genes via the chain rule:

$$\Delta \mathcal{H}_i^{(t+1)} = \sum_M \frac{\partial \mathcal{H}_i}{\partial M_i} \Delta M_i^{(t)}. \quad (12)$$

When ligand expression changes, received ligand fields are recomputed through the Gaussian kernel (equation 2). Ligand-expression deltas are reconciled with received-ligand changes:

$$\Delta_{i,\ell} \leftarrow \Delta_{i,\ell} + \left(\tilde{L}_{i,\ell}^{(t)} - \tilde{L}_{i,\ell}^{(0)} \right) - \left(y_{i,\ell}^{(t)} - y_{i,\ell}^{\text{base}} \right), \quad (13)$$

967 where $\tilde{L}_{i,\ell}^{(t)}$ is received ligand for cell i and ligand ℓ at iteration t . The $(y_{i,\ell}^{(t)} - y_{i,\ell}^{\text{base}})$ term
 968 term strips out the direct perturbation-induced change in ligand expression at cell i ,
 969 preventing double counting once $\Delta_{i,\ell}$ is updated to reflect spatially diffused received
 970 ligand.

971 Jacobian evaluation, intracellular propagation, and ligand recomputation are repeated
 972 for T steps (default $T = 4$; Supplementary Figs. 9, 10). To resolve the molecular routes
 973 underlying simulated transitions, SpaceTravLR decomposes transcriptomic shifts into
 974 hop-specific regulatory contributions at each propagation step.

975 Let $\mathcal{H}_v$ be a target gene and M_u a modulator. The cumulative gradient $\nabla_{\text{cum},v,u,i}^{(n)}$ at step n
 976 is subtracted to obtain the isolated contribution at that hop:

$$977 \quad \nabla_{v,u,i}^{(n)} = \begin{cases} \nabla_{\text{cum},v,u,i}^{(n)} & \text{if } n = 1 \\ \nabla_{\text{cum},v,u,i}^{(n)} - \nabla_{\text{cum},v,u,i}^{(n-1)} & \text{if } n > 1. \end{cases} \quad (14)$$

978

979 Gradients are averaged across cells to construct a directed weighted graph $G = (V, E)$
 980 with edge weights

$$981 \quad w_{uv} = \frac{1}{N} \sum_{i=1}^N \nabla_{v,u,i}^{(n)}. \quad (15)$$

982 Edges with $|w_{uv}| < 10^{-4}$ are pruned.

983 Given perturbation source s and downstream target t , SpaceTravLR identifies influential
 984 regulators within graph distance k by greedy backward tracing from direct regulators of t
 985 to s , selecting maximum-weight in-neighbors at each step. Requiring distinct penultimate
 986 nodes yields diverse parallel pathways rather than redundant variants of a single
 987 cascade.

988

989 **Transition probabilities**

990 The probability that cell i shifts toward the expression state of neighbor j is defined by
 991 Pearson correlation of the simulated shift δ_i with the pairwise expression difference
 992 $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i$, followed by softmax over candidate neighbors $\mathcal{N}_i$ (typically k -nearest
 993 neighbors in UMAP space):

$$994 \quad p_{ij} = \frac{\exp(\text{corr}(\delta_i, \mathbf{r}_{ij})/\tau)}{\sum_{j' \in \mathcal{N}_i} \exp(\text{corr}(\delta_i, \mathbf{r}_{ij'})/\tau)}, \quad (16)$$

995 where τ is a temperature parameter.

To remove embedding-intrinsic background structure, a null transition matrix P_{null} is computed under zero perturbation shift and subtracted ($P_{\text{calibrated}} = P - P_{\text{null}}$).

Vector field functional alignment

Transition probabilities are projected onto a low-dimensional embedding to obtain expected displacement $\mathbf{V}_{i,\text{projected}} = \sum_j p_{ij} \mathbf{V}_{ij}$ per cell. A smoothed vector field is computed following the Velocityto¹²⁷ framework:

$$v_{\text{grid}} = \sum_{i \in F} K_{\sigma}(g, V_{i,\text{projected}}) \cdot V_{i,\text{projected}}, \quad (17)$$

where g indexes grid points and F the neighboring cells, with smoothing kernel $K_{\sigma}(a, b) = \exp(-\|a - b\|_2^2 / (2\sigma^2))$. Functional concordance with an independent reference score (pseudotime, gene-set activity, or related metric) is quantified by the inner product or cosine similarity between simulated shift vectors and reference gradients; positive, negative, and zero values indicate aligned promotion, inhibition, and independence, respectively.

Embedding-based transcriptomic perturbation estimation

We introduce a complementary method for estimating transcriptomic responses to gene perturbations by leveraging lower-dimensional embeddings that represent cellular transcriptional states. Let M be an embedding function (e.g., a trained autoencoder) and G^0 the baseline gene expression matrix. To simulate a perturbation, a modified matrix G^{pert} is created by setting the perturbed gene(s) to target expression value(s). Both G^0 and G^{pert} are passed through M , yielding embeddings whose difference defines the perturbation effect: $\Delta E = M(G^{\text{pert}}) - M(G^0)$. Transition probabilities can then be computed from ΔE as described earlier.

We compared SpaceTravLR against different state-of-the-art models ranging from scFoundation models to simple linear models as well as a custom spatial baseline. For TahoeX1, CellPLM and SpaceFlow, we used the generated embeddings and pipeline described above while for CellOracle we used the predicted gene expression matrix similar to SpaceTravLR. To assess the downstream gene expression changes from CellPLM embeddings, we processed the generated $M(G^{\text{pert}})$ embeddings through CellPLM's decoder module to reconstruct the cell x gene expression matrix.

Since no existing method can predict how one cell influences its immediate neighbors, we develop our own baseline model. This *SpatialBaseline* model uses a graph neural network architecture to predict the gene expression of cells within 200 microns of a given

target cell. During training, we minimize the root mean square error of predicting the gene expression profile of the 50 nearest neighbors given the gene expression matrix of a randomly chosen cell.

Computing differentiation alignment scores (DAS)

To quantitatively assess the impact of different perturbations on different cell types, we define a cell type specific score to measure the alignment between the differentiation trajectory vector and the simulated perturbation vector. We calculate the differentiation vector field using the DPT (*scanpy.tl.dpt*). These single cell pseudotime values are then projected onto the same grid coordinates used for perturbation vector fields via kNN regression. The final cosine similarity between the perturbation vector field $\mathbf{v}_{\text{spacetravlr}}$ and the differentiation (pseudotime gradient) vector field $\mathbf{v}_{\text{pseudotime}}$ at each grid point g is computed as

$$c\text{-DAS}_g = \frac{\mathbf{v}_{\text{spacetravlr}}(g) \cdot \mathbf{v}_{\text{pseudotime}}(g)}{\|\mathbf{v}_{\text{spacetravlr}}(g)\| \|\mathbf{v}_{\text{pseudotime}}(g)\|} \quad (18)$$

where “ $\cdot$ ” denotes the dot product and $\|\cdot\|$ is the Euclidean norm and c is a specific cell type.

For example, Tfh-DAS measures whether a perturbation would promote, inhibit, or be independent of the inferred differentiation axis for T follicular helper cells.

SpaceTravLR also generates two randomized regulatory networks as a negative control. Specifically, we generate fully randomized network edges with random signs and random magnitudes within the same range as the non-random network, and we permute the spatial coefficients per gene such that the overall distribution is the same but the spatial information is lost. We then calculated threshold values for the negative DAS analysis in the systematic KO simulation. The score distribution generated from the randomized controls are used as a null distribution for two-tailed Wilcoxon tests comparing the SpaceTravLR simulated vectors versus the negative controls.

Non cell-autonomous effects of perturbations

By estimating the local changes in ligand received Δ_l by different cells at each location SpaceTravLR is able to accurately predict how a particular cell autonomous perturbations induces changes in gene expression in neighboring cells. To quantitatively assess the differentially expressed genes in the neighboring cells, we first build a kDTree for each sample to efficiently find the k nearest neighbors within a 200 micron radius around cells that were directly perturbed identified by the guide RNAs.

In the murine lymph node, since the 1DER-specific T cell population cannot be isolated with absolute precision, we utilized the broader Th2 cell compartment as a proxy. For control cells, we selected neighbors around cells that received no perturbations.

For the perturb-seq and MERFISH data, we also ensured that these cells were over 500 microns from any cell that received any perturbation. This ensures that the control cells are in an unperturbed environment. DEG analysis between the control neighbors and perturbed neighbors were then done using `scanpy.tl.rank_genes_groups`.

Finally, to evaluate the performance of SpaceTravLR, we trained the model on a wild type single cell ST dataset from the same tissue. For in silico perturbations, we selected 300 cells of the same cell type and in the same anatomical region as the cells that were perturbed in the experimental dataset. We then repeated the same process to find neighbors around these cells, and around control unperturbed cells in the same region but in unperturbed environments. To quantify the concordance between SpaceTravLR's predicted DEGs and the experimental DEGs, we report both the spearman and pearson correlations.

Ranking functional cell-cell interactions

Ligand-Receptor signaling activity defined by Equation 1 varies spatially and the modulatory effect of those LR interactions depends on the cell type, the receptor expression on individual cells and the local ligand concentration around each cell. SpaceTravLR explicitly models the interaction strength at each location and the induced downstream gene expression changes across the whole transcriptome. This allows us to compute the niche specific impact of different LR interactions on a group of genes. For example, to estimate the functional impact of a particular ligand-receptor pair LR on a set of gene S , we aggregate all the coefficients for each LR and weigh the interactions by the receptor expressed on the particular cells of interest

$$\frac{\partial S}{\partial LR} = \sum_{g \in \text{genes}} \frac{\partial g}{\partial LR} R$$

1091 We can repeat this process for multiple LR pairs to rank the ligand-receptor pairs driving
1092 the gene signature in a specific direction (higher or lower).

1093

1094 **Shapley analysis**

1095 To quantify the relative contribution of each SpaceTravLR feature family, we performed
1096 an exact Shapley⁶⁷ decomposition. We treat TF, LR, LTF and Spatial as the four
1097 contributors.

1098 Let the player set be

$$1099 \quad N = \{\text{TF, LR, LTF, Spatial}\}, \quad |N| = 4. \quad (19)$$

1100 For every coalition $S \subseteq N$, we trained one SpaceTravLR model with only the families in S
1101 enabled.

1102 The contribution of feature family $i \in N$ to gene g is

$$1103 \quad \varphi_i(g) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]. \quad (20)$$

1104

1105 **Identifying microniches from spatial transcriptomics**

1106 For broader, non-overlapping niche partitions, regulatory activity is summarized across
1107 highly variable genes. The modulator activity of modulator m at cell i is

$$1108 \quad a_{im} = \frac{1}{|G_{HVG}|} \sum_{g \in G_{HVG}} \beta_i^{m,g}, \quad (21)$$

1109 where $\beta_i^{m,g}$ is the SWR coefficient linking modulator m to gene g . The resulting matrix
1110 $A \in \mathbb{R}^{N \times M}$ is embedded via a k -nearest-neighbor graph and partitioned with Leiden
1111 clustering to obtain discrete functional microniches.

1112

1113 **Perturbation induced microniches**

1114 In this analysis, SpaceTravLR was trained on sub-sections of the MC38 and Lung
1115 metastasis VisiumHD datasets excluding the cells that received any targeting guides and
1116 cells within a 400 micron radius in order to also exclude indirectly perturbed cells.
1117 Observed enrichment per niche was summarized as the $\log_2$ odds ratio of sgP-to-NTC
1118 cell fractions, similar to TARDIS¹⁰. SpaceTravLR predicted niche favorability on the same

cells from (i) a baseline immune-exclusion index, (ii) in-silico KO escape gain, combined after z-scoring with perturbation-specific weights. Microniches were defined on the unperturbed tumor pool via Leiden clustering on the learned spatial coefficients from the vision model. Leiden parameters were tuned to maximize the silhouette score computed using `sklearn.metrics.silhouette_score`. Concordance was assessed as Pearson r across microniches for each slice-perturbation pair, with negative controls from permuted labels.

Labeling known tissue architectures

Within the tonsil germinal centers, there are three distinct regions: the dark zone (DZ), light zone (LZ), and gray zone (GZ), each identified by a distinct transcriptomic profile. The DZ is identified by *Cxcr4*, *Aicda*, *Foxp1*, and *Mme*, the LZ more highly expresses *Cd83* and *Lmo2*, and the GZ expresses a combination of *Top2a*, *Cenpp*, and *Cd83*. Thus, we use Leiden clustering on the GC B cells and categorize most of the clusters into DZ if they are $Cxcr4^{high}Cd83^{low}$, LZ are $Cxcr4^{low}Cd83^{high}$, and the GZ are $Cxcr4^{high}Cd83^{high}$. The cells in the unassigned clusters were subsequently annotated as LZ, DZ, GZ, or non-GC based on the majority cell-type fraction in their neighborhoods (10 nearest GC B cell neighbors within 200 μm). In this manner, we classified 1152 LZ, 473 DZ, 213 GZ, and only 10 non-GC.

Within the murine lymph nodes, there are also distinct regions: the T cell zone, TB-border, and B-cell follicles. For each cell, we computed the proportion of B cells and T cells (B/T) within their neighborhoods (100 nearest neighbors). Spatial zonation was determined by comparing the local B/T ratio against the global tissue baseline. Neighborhoods maintaining a local ratio within $\pm 2/3$ of the global baseline were classified as the TB border. Ratios exceeding this threshold designated B-cell follicles, while ratios falling below it identified T-cell zones.

Impact of perturbations on niche preference

To quantitatively assess the shift in niche preference of a cell following a perturbation, we use a Monte Carlo Permutation approach to generate a background distribution representing "random chance" of a cell associating with a given tissue region. We randomly select cells from the entire dataset, find their nearest spatial neighbors and then calculate the observed proportions, repeating this process 1000 times.

$$\text{EnrichmentScore} = \frac{\text{observed proportion}}{\text{expected proportion}} \quad (22)$$

$$Z = \frac{\text{observed} - \text{expected}}{\text{standard deviation}} \quad (23)$$

To compare two samples (control vs ko), we normalize for the fact that different tissues have different proportions in different zones. Therefore, we calculate a separate enrichment score (Observed/Expected) for each sample.

$$\log_2 \text{fc} = \log_2 \frac{\text{EnrichmentScore (KO)}}{\text{EnrichmentScore (control)}} \quad (24)$$

Here, a positive log2fc for a given zone indicates that *Ccr4* KO Th2 cells moved closer to that zone, a negative log2fc indicates that *Ccr4* KO Th2 cells moved away from that zone and a zero Log2fc indicates no the spatial preference between the two samples.

***Patz1* KO multivariate analysis**

Flow cytometry data from the *Patz1* KO adoptive transfer experiments were analyzed at the mouse-level with canonical pMEL and host CD8 T cell marker readouts. We compared *Patz1* KO to scramble treated pMEL cells from mice adoptively transferred with either 5×10^4 or 1×10^5 cells from the independent experiments analyzed at days 17-19 post tumor inoculation. A total of 24 mice (4 mice per group) across all timepoints and experiments were assessed (as the 5×10^4 *Patz1* condition was excluded). To control for scale differences, marker expression was z-scored (standardized). The joint *Patz1* phenotype was modeled using Linear Discriminant Analysis (LDA) to project marker profiles into a two-dimensional signature space. A linear Support Vector Machine (SVM) was then trained on the LDA coordinates to classify *Patz1* KO versus Scramble control mice. Statistical significance of this classification was assessed via 1,000 Monte Carlo label permutations using Leave-One-Out cross-validation.

SpaceTravLR implementation

The new Rust implementation turns SpaceTravLR from a small research Python tool into a production software you can install with one command, without juggling conda environments or CUDA drivers. We reimplemented the full pipeline from preprocessing, sparse group lasso fitting, CNN training, and perturbation propagation in pure native rust with extensive parallelization that isn't possible using Python. All tensor-heavy steps such as computing the spatially weighted received-ligand, generating the spatial proximity grids, lasso FISTA regression, and multi-gene training are now orders of magnitude faster and scale to millions of cells.

Experiments

All work with mice was done in accordance with the Institutional Animal Care and Use Committee at the University of Pittsburgh.

***Ccr4* KO**

C57BL/6J (0000664) were purchased from Jackson Laboratories. 1DER mice were previously described. All animals were housed in specific pathogen-free conditions in the Assembly building. All experiments were approved by the University of Pittsburgh Institutional Animal Care and Use Committee. All mice were over 6 weeks of age and both male and female mice were used. Naive CD4 T cells were enriched from 1DER TCR Tg mice using Naive CD4 T cell isolation kit (STEMCELL,19765). For *Ccr4* KO by CRISPR-Cas9, naive CD4 T cells were electroporated with murine target or scramble control Single-guide RNA (sgRNA). The detailed protocol using the CRISPR-Cas9 system was previously described¹¹⁶. sgRNA targeting murine *Ccr4* and scramble control sgRNAs were obtained from Synthego (CRISPR Gene Knockout Kit v2). The Cas9 nuclease V3 was obtained from Integrated DNA Technologies (1081059). The sgRNA-Cas9 RNA complex was introduced into naive CD4 T cells using Lonza 4D-Nucleofector Core Unit (Lonza, AAF-1002B). All deletions were validated by flow. After CRISPR/Cas9 deletion, a total of 1×10^6 CD4 T cells were injected into congenic recipients. After 24h, mice were immunized with HDM (25 μ g of lipopolysaccharide-low HDM (Stallergenes Greer) in PBS) intranasal to mice anesthetized with isoflurane daily for 3 days. Mice were sacrificed on day 3, mediastinal lymph nodes harvested and fixed in formalin for paraffin embedding.

***Patz1* KO**

C57BL/6, pMEL and *Rag1*^{-/-} mice were purchased from the Jackson Laboratory. All animal experiments were performed in the American Association for the Accreditation of Laboratory Animal Care-accredited, specific-pathogen-free facilities in the Division of Laboratory Animal Resources at the University of Pittsburgh School of Medicine. Only 6–8-week-old mice were used for studies. Animal protocols were approved by the Institutional Animal Care and Use Committees of the University of Pittsburgh.

CRISPR-Cas9-mediated knockout

Guide RNAs (gRNA) for *Patz1* (CGGGTCAACGACGCTTCTTG and TCTCTAGGCGCACAACGATT) and a nontargeting scramble control were purchased from Integrated DNA Technologies (IDT). Scramble, *Patz1*, and *Pdcd1* ribonucleoproteins (RNPs) were generated by first annealing crRNAs with tracrRNA to form sgRNA

complexes (IDT) and then combining with Cas9 nuclease (IDT) using commercial guidelines (IDT). Naïve pMEL *Rag1*^{-/-} CD8⁺ T cells were isolated and stimulated for 36 hours using plate bound anti-CD3 (Biolegend; 2.8 µg/mL), soluble anti-CD28 (Biolegend; 2.8 µg/mL), and IL-2 (100 U/mL) in complete RPMI 1640 (Invitrogen). After pre-stimulation, cells were electroporated with *Patz1*, *Pdcd1*, or scramble RNPs using the P4 Primary Cell 4D-Nucleofector Kit (Lonza) with setting CM137 on the 4D-Nucleofector System (Lonza). Following electroporation, cells were transferred to pre-warmed complete RPMI 1640 and rested for 2 hours at 37 °C. Cells were centrifuged and resuspended in complete RPMI 1640 supplemented with IL-2 (100 U/mL) and IL-7 (10 ng/mL) and incubated at 37 °C. After 24 hours, 5 x10⁴ or 1 x10⁵ *Patz1*-targeted, *Pdcd1*-targeted, or scramble-targeted pMEL cells were adoptively transferred into recipient mice by intravenous tail vein injection.

Tumor Models

All mice were intradermally injected with 1.25 x10⁵ B16-gp100 cells one day after adoptive transfer of pMEL cells. The operator was blinded to the groups. Blinded tumor measurements were taken every 3 days with digital calipers and tumor areas were calculated. Tumors were collected for analysis between days 17-19 post tumor injection. Tumors were mechanically disrupted and enzymatically digested with collagenase (Sigma; 1 mg/mL) and DNase (Roche; 20 µg/mL) in complete RPMI 1640 at 37 °C for 30 minutes. Samples were dissociated through a 70 µm filter, washed with separation buffer (PBS, 2% fetal calf serum, and 2mM EDTA) and resuspended in 5 mL of 40% Percoll (Cytiva) with a 1 mL 80% Percoll underlay – 100% Percoll solution was made by diluting Percoll in 10X PBS at a ratio of 9:1, before diluting in RPMI 1640. Following centrifugation, lymphocytes were isolated from the middle layer of the Percoll gradient, washed, and stained for flow cytometric analysis.

Single-cell suspensions were stained with anti-mouse antibodies against FAS (15A7; eBioscience), ITGB (eBioHMb 1-1; eBioscience), CD45.2 (104; Biolegend), CD47 (mlap301; eBioscience), PD1 (RMP1-30; eBioscience); CD90.1 (OX-7; BD Biosciences), CD8 (53-6.7; eBioscience), and LAG3 (410C9; in-house). Dead cells were discriminated using Ghost Viability Dye Red 780 (Cytek) in PBS. Cells were blocked with normal mouse serum for 15 minutes at 4°C followed by surface staining for 30 minutes on ice.

Ccr5 KO

For *Ccr5* knockout (KO) studies, P14 T cells were isolated using the EasySep Mouse CD8⁺ T cell isolation kit (Stemcell 19853). Enriched P14 T cells were mixed with either three pooled *Ccr5* sgRNAs (Synthego sgRNA EZ Kit; UCAGUCUAUACCCGAUCCAC,

UAGUUCUAAUAGACUAGACC, UAUAGCUAUGACAUCGAUUA) or sgCD19 sgRNA (IDT; GGUGGAAUGCUUCAGACGUC) and Cas9-GFP (IDT) complexes, and electroporated using the P3 Primary Cell 4D-Nucleofector X kit and unit (Lonza, Cat# V4XP-3032). Electroporated cells were rested for 30 minutes at 37°C, then mixed at a 1:1 ratio and 1×10^5 total cells were transferred into naïve B6 recipient animals and infected with LCMV Armstrong 24 hours later. To validate *Ccr5* KO, a subset of electroporated cells were activated in vitro with plate bound goat-anti-hamster IgG (100 μ g/mL, H+L, Thermo Fisher Scientific) with 1 μ g/mL anti-CD3 and 1 μ g/mL anti-CD28 (eBioscience). After 24 hours, 10 ng/mL of IL-12 (BioLegend) and 10 ng/mL of IL-2 (PeproTech) were added to the culture. *Ccr5* KO was confirmed 48 hours after electroporation by flow cytometry. Seven days after LCMV Armstrong infection, spleen and small intestine tissues were harvested and single cell preparations were prepared. Small intestine intraepithelial lymphocytes (siIEL) were prepared by excision of Peyer's patches and removal of luminal contents. Then the entire tissue section was cut longitudinally, then cut laterally into 1 cm pieces, which were then incubated at 37°C for 30 minutes in Hanks' balanced salt solution with 2.4 mg ml⁻¹ of Hepes, 2.1 mg ml⁻¹ of sodium bicarbonate, 8% bovine growth serum and 0.154 mg ml⁻¹ of DTE (EMD Millipore). Lymphocytes from non-lymphoid tissues were further enriched using a 44%/67% Percoll density gradient. Single cell suspensions of spleens were prepared by mechanical disruption, followed by ACK-lysing buffer.

Single-cell suspensions were stained with the following antibodies: BUV395 CD45.2 (BD, #564616), BUV496 CD103 (BD, #741083), CD8b BUV805 (BD, #741992), CD69 BV421 (Biolegend, #104545), CD45.1 BV785 (Biolegend, #110743), PE CCR5 (Biolegend, #107006), PE-Cy7 P2XR7 (Biolegend, #148708), APC IL18R (Biolegend, #157905), APC eF780 I.V. CD8a (eBioscience, #47-0081-82) and Fixable Viability Dye eFluor 506 (eBioscience, #65-0866-18) and analyzed on the BD Symphony.

XYZeqV2 murine kidney experiments

Kidneys from 8-week-old C57BL/6 female mice were purchased from Envigo Bioproducts and maintained in RPMI 1640 media. The kidney organ was mounted on a microtome (Precisionary Compresstome) and sliced at 300 μ m to use on the Cytope microwell array. The microwell array chips were stored at 4°C and equilibrated to room temperature before mounting the kidney tissue slice on top of the array. A digital image was taken to record the tissue orientation after the tissue was pressed onto the Cytope array, secured between the chip and a glass microscope slide. The chip was placed inside a humidity chamber at room temperature and incubated for 30 minutes. After the incubation period, the chip was removed from the humidity chamber and immediately washed in an initial reagent reservoir containing 40ml of RPMI with 10% FBS which was followed by a second

wash in a separate reagent reservoir containing 25ml of PBS to rapidly dilute and remove any unbound spatial label. The tissue slices were transferred into their respective sample tubes containing 0.5ml of 2kU/ml Collagenase IV with protease inhibitor (Millipore Sigma C6079) and dissociated at 37°C for 30 minutes with trituration every 10 minutes. The cells were filtered through a 70 μ m cell strainer and washed once with RPMI and 10% FBS. To ensure fluorescence activated cell sorting (FACS) of only the cells that are labeled with the printed antibodies from the array, the cells were stained with 1 μ l of PE goat anti-rat polyclonal secondary antibody (Biolegend 405406) in 100 μ l of PBS for 30 minutes on ice. The cells were washed twice, with 2ml of PBS and 10% FBS, and resuspended in a final volume of 0.5ml of 25nM Helix NP NIR (Biolegend 425301). All samples were strained through a 35 μ m filter before FACS.

To acquire high quality cells for downstream processing, cells characterized as smaller than 4 μ m and doublets were gated out. Cells that were negatively stained for the viability stain, Helix NP NIR, were gated into a FSC v PE channel plot where cells that were positively labeled for the goat antirat polyclonal secondary were sorted into a 1.5ml Eppendorf tube coated with heat inactivated FBS.

Sorted cells were centrifuged and resuspended in phosphate-buffered saline at 1000 cells/ μ l and loaded on the 10x Genomics Chromium platform and processed using the Next GEM Single Cell 3' with Feature Barcoding protocol for library preparation. Sample libraries were sequenced on the Novaseq X.

Data availability

All datasets generated and used in this study is freely accessible and the preprocessed datasets have been archived for reproducibility. We downloaded human tonsil SCP2169, SCP2171, and SCP2176 from the Broad Institute Single Cell Portal (https://singlecell.broadinstitute.org/single_cell). We also downloaded murine brain datasets under the accession number SCP815. We downloaded [human lymph node](#)⁵³, [Human cervical cancer](#)¹⁰⁷ and [human breast cancer](#)¹⁰⁸ from the 10x website. We downloaded spatial perturb-seq murine hippocampus Stereoseq datasets from the Gene Expression Omnibus (GEO) with accession number GSE274447⁸⁴ and the murine embryo from the Chan Zuckerberg CELLXGENE portal, which is also available under the accession number GSE197353. Both murine small intestine datasets were from previously published⁵⁷ and are available at GSE280895 and GSE279254. We had also previously published the murine lymph node dataset at GSE245112¹¹⁵. The experimental spatial perturb-seq data of murine liver was downloaded from GSE275483⁸⁵, the murine hippocampus from GSE274447⁸ and the murine tumor from the [SPAC-seq](#) official web portal. The murine skin dataset⁶³ was requested from collaborators in advance of their preprint and will be made available once published. The XYZeqV2 murine kidney datasets

were generated and annotated in collaboration with SurveyGenomics. All spatial transcriptomics datasets generated for our study (two mouse kidneys at true single-cell resolution generated by Survey using novel technology, one VisiumHD mouse lymph node in the allergic HDM model, and two VisiumHD mouse lymph nodes with knockout adoptively transferred cells) will be available at time of publication.

Code availability

The original python implementation of SpaceTravLR (including all the code, data, documentation and examples) are available on GitHub at <https://github.com/jishnu-lab/SpaceTravLR>. The new rust implementation is available at https://github.com/Koushul/SpaceTravLR_rust and the tutorial at <https://spacetravlr-rust.readthedocs.io/en/latest/usage/>. We recommend using the rust version which can be install with the command ``curl -fsSL https://tinyurl.com/spacetravlr/scripts/install.sh | sh``

Acknowledgements

J.D. was supported in part by NIAID DP2AI164325, NIAID R01AI170108, NIAID U01AI179514 and NHGRI U01HG012041. Y .L. has received funds from the Parker Institute for Cancer Immunotherapy (PICI). A.C.P. was supported in part by NIAID R01 AI153104 and NIAID R01 AI156093. D.A.A.V. was supported by NIAID P01 AI108545 and NCI R35 CA263850, and A.B. by NIH F32 CA298301. The authors acknowledge the resources and services provided by the Rangos Histology Core Facility, a shared facility in the Rangos Research Center at the University of Pittsburgh. This project utilized the services of the University of Pittsburgh Health Sciences Sequencing Core at UPMC Children's Hospital of Pittsburgh to perform VisiumHD. This research was supported in part by the University of Pittsburgh Center for Research Computing and Data, RRID:SCR_022735, through the computing resources provided. Specifically, this work used the H2P cluster, which is supported by NSF award number OAC-2117681 and the HTC cluster, which is supported by NIH award number S10OD028483.

Contributions

J.D. conceived the initial idea of the project. K.R. and A.W. designed, implemented, and benchmarked all components of the SpaceTravLR algorithm. S.G. and W.M generated the *Ccr4* KO and WT VisiumHD datasets under the supervision of A.P.. A.B. and C.I. generated *Patz1* KO data under the supervision of C.W. and D.V.. K.T. generated the *Ccr5* KO data under the supervision of M.H.. H.L generated the XZyEqV2 murine kidney datasets. J.C conducted scRNAseq data analyses for melanoma. K.R. and A.W. preprocessed and analyzed all the datasets. Y.C.L. conducted scATACseq and

CellOracle analyses for melanoma in-silico screens. C.S. and D.K provided Xenium skin samples and provided feedback on analyses. Y.C. helped refine the annotations for the Slideseqv2 murine lymph node dataset. J.S. tested SpaceTravLR and provided feedback on cross-sample comparisons. N.L provided feedback on analyses. K.R., A.W. and J.D. wrote the manuscript and generated the figures and illustrations with contributions from all authors. J.D., A.P., Y.L. supervised the project. All authors read and approved the final manuscript.

Ethics declarations

Competing interests

Y.L. is a cofounder of Survey Genomics. D.A.A.V. declares the following interests: Novasenta/AI3bio; stock holder, Trishula; stock holder – Werewolf; patents licensed and royalties - BMS, Novasenta; scientific advisory board member - Werewolf, Apeximmune, T7/Imreg Bio, Secarna; consultant - Third Arc Bio, Curie Bio, Ablytix, Radionetics Oncology, Ellipses; funding - BMS, Novasenta.

The remaining authors declare no competing interests.

Figures

Figure 1: Linking spatial and functional biology with SpaceTravLR.

A. SpaceTravLR processes a spatial transcriptomics dataset as input for a CNN to capture localized spatial context (top left). The network estimates spatially informed cell-specific coefficients for three distinct classes of modulator terms: TFs, ligand-receptor pairs, and ligand-transcription factor associations (bottom). Utilizing these learned coefficients and signaling variables, SpaceTravLR applies chain rule to simultaneously simulate both intrinsic and extrinsic cellular perturbations (top right). While the schematic depicts a single, linear pathway propagating the effect of a single gene, the framework simultaneously computes the derivatives of the entire system, resolving all cellular responses to all perturbation variables through parallel tensor operations for each iteration (Methods).

B. Comparison of the predictive capabilities of state-of-the-art methods against SpaceTravLR. CellPLM and Tahoe-x1 utilize cell-specific embeddings to illustrate the cell-intrinsic effects of a given perturbation. SpaceFlow incorporates localized spatial information to generate spatial embeddings for cell-intrinsic predictions, while CellOracle constructs gene regulatory networks (GRNs) restricted to cell-intrinsic dynamics. In contrast, SpaceTravLR leverages spatial context to simultaneously visualize and predict both cell-intrinsic and cell-extrinsic downstream effects of perturbation.

C. SpaceTravLR's unique ability to propagate the effects of perturbation beyond the target cell enables the modeling of higher-order spatial architecture reprogramming.

D. SpaceTravLR links space with function by uncovering regions with varying responses to specific gene perturbations due to disparities in their underlying modulator activities. Downstream analysis of these *in silico* simulations enables systematic investigation into the localized impacts of cell-cell communication, facilitating the discovery and prioritization of candidate gene targets for experimental validation.

Figure 2: SpaceTravLR recapitulates the expected transcriptomic outcomes and spatial dependencies in diverse cell types within a complex tissue environment.

A. Spatial organization and UMAP projection of the complex cell types in the Slide-tags human tonsil dataset with distinct germinal center structures. Schematic illustrates the current biological understanding of key regulatory genes in these cell types and their interactions.

B. Quantification of transcriptomic shifts following gene perturbations in distinct subpopulations of cells. Dot plots illustrating the mean shift in transcriptome across diverse cell lineages. CXCL13 OE within the FDC population selectively drives shifts in neighboring, CXCR5-expressing populations. Similarly, lineage-specific OE of IL21 in Tfh cells and IL18 in myeloid cells selectively alters adjacent cells expressing their corresponding receptors IL21R and IL18R, respectively. A BCL6 KO confined entirely to Tfh cells triggers significant transcriptomic shifts exclusively within nearby IL21R expressing populations.

C. Vector fields overlaid on the UMAP layouts, subset to CD4 T cells, showing the direction and magnitude of transcriptomic changes following single-factor or combinatorial perturbations. To evaluate directional accuracy, the simulated shifts are compared against three distinct pseudotime trajectories constructed to map alternative CD4 T cell differentiation routes: from Th0 to Th1, Th0 to Tfh, and Th0 to Th2. Cosine similarity was used to measure the direction of cell fate shift resulting from each perturbation. These alignments demonstrate that natively simulated Bcl6 KO or Il21 KO systematically shifts the neighborhood vector field to encourage the adoption of a Th1 cell fate. Conversely, Bcl6 OE or Il21 OE drives cellular trajectories toward a Tfh cell fate. Combinatorial perturbations chart overlapping and epistatic neighborhood-wide consequences, mapping how cooperative direct and indirect signaling networks dictate local cell fate bias.

E-F. Scatter plots evaluating the predicted CellPLM (left) and SpaceTravLR (right) log2fc change of target genes CXCR5 (**E**) and BCL6 (**F**) relative to spatial proximity to GC B cells resulting from IL21 OE in naive CD4 T cells and colored by STAT3 log2fc. Although CellPLM incorporates spatial metadata in its embeddings, it fails to capture the distance dependent relationship between gene expression dynamics and the density of GC B cells in the local environment, failing to predict the directionality of the expected differential expression. SpaceTravLR uniquely incorporates spatial configurations to accurately capture continuous, distance-dependent extrinsic perturbation effects, exhibiting a strong positive Spearman correlation (0.4887 for PDCD1 and 0.4795 for CXCR5).

Figure 3: SpaceTravLR captures spatial cell-to-cell variability in perturbation response within germinal centers.

A. Simplified subnetwork of *Foxo1* previously established in the literature. *Foxo1* positively regulates the expression of *Batf* and *Cxcr4* which are essential for the expression of *Aicda*, an essential gene for somatic hypermutation in GC dark zones. *Foxo1* has also been linked to the survival of naive CD4 T cells while its deletion has been confirmed to amplify Tfh populations.

B. Representation of the spatial architecture of the germinal centers consisting of the dark zone (DZ), gray zone (GZ), and the light zone (LZ) (left). *In-silico* *Foxo1* KO induces the ablation of the dark zone and intermediate zone regions (right).

C. Heatmap illustrating the significance of predicted state transitions toward the LZ compartment following *Foxo1* disruption. Permutation testing via post-perturbation gene shuffling demonstrates that SpaceTravLR's prediction of cells adopting an LZ fate is statistically significant against a randomized null distribution.

D. Ranking of top LR interactions driving AICDA expression in DZ GC B cells. Cell types displayed as ligand source are broader groupings of the annotations to improve visibility.

E. SpaceTravLR and **F.** CellOracle predicted log2 fold change of key GC genes following *Foxo1* KO. From existing literature, *Bcl6* expression is expected to increase while *Batf*, *Cxcr4*, and *Aicda* are expected to decrease. *Trav6* is a T-cell receptor gene that is not biologically relevant in GC B cells. Similarly, as a negative control, *C2orf50* encodes a microtubule inner protein essential for the structural integrity and motility of cilia.

G. Heatmap illustrating the significance of predicted state transitions toward the LZ compartment following *Foxo1* disruption as simulated by CellOracle. Permutation testing via post-perturbation gene shuffling demonstrates that CellOracle's prediction of cells adopting an LZ fate is statistically significant against a randomized null distribution. However, DZ cells are also statistically significantly likely to retain a DZ fate.

H. Schematic of the cross-tissue transfer. To apply our model trained on the higher-resolution Slide-tags tonsil (top left), to the lower-resolution murine lymph sample profiled with 10x Visium (top right), we align neighborhoods consisting of 5 single cells from the tonsil to each 55um Visium spot, which may contain anywhere from 1 to 10 cells. For each test spot, we identify the neighborhood in the training sample with the most similar structure as the cell-type abundance scores based on Pearson correlation. The final coefficients used by the cells in the test sample are the averages of the cell-level coefficients within the matched neighborhood

I. Spatial layout of a Visium human lymph node sample, where each spot contains multi-cellular transcriptomic profiles. Cell2Location was utilized to assign cell-type abundance scores to each spot. Non-GC spots are dominated by alternative cell-types and are computationally masked in gray to better visualize the GC areas. The continuous color gradient maps the localized intensity of the assigned Cell2Location DZ GC B cell score.

J. In-silico *Foxo1* perturbation on the Visium human lymph node using the modulator coefficients derived from the Slidetags human tonsil caused a pronounced transcriptomic shift directed away from microenvironments exhibiting high Cell2Location DZ abundance scores.

K. Spatial layout of human tonsil CD4 T cells partitioned by Tfh and non-Tfh identity under baseline homeostatic conditions (left). Targeted in silico *Foxo1* KO drives a strong

directional cell fate bias that results in a severe predominance of the Tfh population (right).

L. Permutation testing, performed via post-perturbation gene shuffling, demonstrates that the shift toward a Tfh fate bias is statistically significant against a randomized null distribution.

M. SpaceTravLR predicted log2 fold change of key CD4 T cell genes following *Foxo1* KO. From existing literature, *Bcl6*, *CXCR5*, and *PDCD1* expression is expected to increase in Tfh cells while *Sell*, *Ccr7*, and *Il7r* are expected to decrease in naive CD4 T cells. *Lyz*, which encodes lysozyme, and *C2orf50* are not expected to change.

Figure 4: SpaceTravLR infers the functional consequences of cell-cell communication events at true single cell resolution in murine kidney.

A. Schematic of paracrine signaling between epithelial cells and macrophages. Epithelial cells secrete *Mif*, which promotes the hypertonic macrophage population.

B. Hypertonic macrophages localize to the inner regions of the kidney, while regular macrophages mostly populate the outer regions.

C. UMAP of kidney single-cell data colored by cell type (top), *Mif* expression intensity (middle), and *Cd74*, one of the receptors for *Mif* (bottom).

D. UMAP streamline projection illustrating continuous cell-state transitions within the macrophage compartment following simulated epithelial cell *Mif* KO. Trajectory vector alignments are quantified by computing the cosine similarity of the neighborhood expression gradients against a trajectory derived from the hypertonic macrophage marker gene signature. Cosine similarity of -0.24 indicates a significant shift away from the hypertonic macrophage identity.

E. Cross-platform validation approach. The model trained on the spatial context of one XYSeqV2 sample is generalized to a separate XYSeqV2 dataset to predict the modulator coefficients.

F. Continuous cell-state transitions within the macrophage compartment following simulated epithelial cell *Mif* KO on a second XYSeqV2 mouse kidney sample. These predictions were generated from another SpaceTravLR model trained directly on this sample (left), and the original SpaceTravLR model trained on the first sample (right). Cosine similarities of -0.44 and -0.41 indicates that both models predict a significant shift away from the hypertonic macrophage identity.

G. The proportion of hypertonic macrophages was quantified under baseline and simulated epithelial *Mif* KO conditions. Prediction consistency was evaluated across three contexts: intra-sample prediction on Sample 1, intra-sample prediction on Sample 2, and cross-sample transfer (trained on Sample 1, evaluated on Sample 2). McNemar's test confirms a statistically significant decrease in the predicted proportion of hypertonic macrophages across all paradigms.

H. Performance comparison of microniches identified by SpaceTravLR trained on native tissue, zero-shot transfer, BANKSY, COVET from ENVI, and Nichecompass. These metrics were computed using Nichecompass's published code and evaluate both niche coherence and spatial consistency.

I-K. SpaceTravLR's global microniche assignment in three contexts: intra-sample prediction on Sample 1, intra-sample prediction on Sample 2, and cross-sample transfer (trained on Sample 1, evaluated on Sample 2). Comparing the spatial niche assignments

generated by the native target model against those generated via cross-sample transfer yields an Adjusted Rand Index (ARI) of 0.612, demonstrating a high degree of architectural consistency across independent tissue sample

Figure 5: Validation of Simulated Non-Cell-Autonomous Perturbation Effects via Spatial Perturb-seq

A. Spatial layout of Stereoseq Spatial Perturb-seq mouse hippocampus section with different guides
B. Close-up spatial layout for neighborhood perturbation analysis comparing neighborhoods around cells receiving sgRNAs vs control cells
C. Spatial layout of Slideseq2 mouse hippocampus section used for training SpaceTravLR
D. Experimental vs SpaceTravLR log2fold change of the significant genes in the local neighborhood of the perturbed cell. Only the combinations of guide type, guide target cell, and neighbor cell type population that have over 20 significant differentially expressed genes in the experimental data and have at least 5 upregulated and 5 downregulated DEGs are displayed here.
E. Spatial layout of MERFISH mouse liver section
F. Spatial layout of Stereoseq mouse liver section used for training SpaceTravLR
G. Experimental vs SpaceTravLR DEGs for neighborhood populations in MERFISH mouse liver

Figure 6: SpaceTravLR Discovers a Novel Gene Target PATZ1 for Immunotherapy Applications

A-B. Spatial layout and UMAP showing annotated cell types in a Slide-Tags human melanoma sample.
C. Gene signature scores overlaid on a UMAP projection of CD8 T cells and tumor cells, delineating microenvironments defined by tumorigenicity (*Pmel*, *Mitf*, *Dct*, *Cdk2*, *Cdh1*, *Tyrp1*, *S100b*), T cell exhaustion (*Tigit*, *Lag3*, *Tox*, *Btla*, *Havcr2*), and T cell activation (*Itga1*, *Gzma*, *Gzmk*, *Ccl5*, *Ccr5*). Directional vector fields were generated utilizing the respective marker gene sets for each category.
D. Tumorigenicity alignment scores from whole transcriptome knockout via SpaceTravLR's in-silico screen
E. Predicted continuous cell-state transitions following targeted genetic perturbations. To quantify directional cell fate bias, simulated transition vectors were evaluated by computing cosine similarity against the above baseline trajectory derived from the marker gene signatures. Individual *PDCD1* and *CTLA4* knockouts resulted in increased activation (A), decreased exhaustion (E), and reduced tumorigenicity (T).
F. Predicted cell-state transitions following individual and combinatorial knockouts of *Prdm1* and *Nr4a3*. The combinatorial knockout decreases tumorigenicity substantially more than either individual perturbation alone.
G. SpaceTravLR identifies a novel hit, *PATZ1*, as a gene candidate whose deletion may result in decreased CD8 T cell exhaustion and decreased tumorigenicity.
H. In-silico deletion of *PATZ1* in CD8 T cells results in altered gene expression in tumor cells. TYRP1% change in tumor cells is negatively correlated with the number of CD8 T cells in its environment

I. Schematic of orthogonal experimental validation. One day prior to tumor challenge, mice receive an intravenous injection of CD8⁺ pMEL T cells, which recognize the gp100 melanoma antigen, from either wild-type, *Lag3*^{-/-}, *Pdcd1*^{-/-}, or *Pdcd1*^{-/-}*Lag3*^{-/-} mice. On the following day, the mice receive an intradermal injection of the B16-gp100 melanoma cell line. Tumors were resected and scRNAseq analysis was performed on day 14.

J-K. Differential gene expression analysis comparing (I) *Pdcd1*^{-/-} and (J) combinatorial *Pdcd1*^{-/-}*Lag3*^{-/-} pMEL cells against wild-type conditions. Both treatments, known to improve tumor outcomes, consistently exhibit elevated *Patz1* expression (red) compared to untreated wild-type populations (blue).

L. Schematic showing experimental validation. Mice receive an intravenous transfer of either wild-type or *Patz1* deficient pMEL T cells followed by an intradermal challenge with B16-gp100 melanoma cells. The tumors were harvested between days 17 and 19 to evaluate how these perturbations impact CD8 T cell exhaustion and activation.

M. Linear Discriminant Analysis on the geometric mean of the mean fluorescence intensity (MFI) of phenotypic markers PDCD1, CD47, FAS, and ITGB1 of the pMEL CD8 cells. Scatter plot shows individual day 17 biological replicates projected onto the first two linear discriminants (LD1 and LD2). Axes represent the linear combinations of markers that maximize separation between the groups, with the percentage of explained between-group variance indicated in parentheses. The dashed black line represents the decision boundary calculated by a linear Support Vector Machine (SVM) trained to classify scramble controls versus *PATZ1*^{-/-}. Cross-validation accuracy (91.7%) and the corresponding significance level ($p=0.025$) calculated via 1,000 Monte Carlo permutations indicate that the *PATZ1*^{-/-} pMEL CD8 T cells are phenotypically different from the scramble controls.

N. Loading weights driving group separation along linear discriminants. Bars represent the relative coefficient weights of each marker for LD1 and LD2. Dark grey bars indicate positive loading weights (associated with higher expression along the positive axis direction), and light grey bars indicate negative loading weights (associated with higher expression along the negative axis direction).

Figure 7: Experimental validation of SpaceTravLR predictions for the spatial relocation of Th2 cells following *Ccr4* knockout in murine lymph node.

A. Schematic of murine model of allergic asthma. CD4 T cells were derived from a T cell receptor (TCR) transgenic mouse specific to Derp-1 (1DER), a class II restricted immunodominant peptide from House Dust Mite (HDM). These cells can be injected directly into the mouse through the tail vein or first altered using CRISPRi to induce *Ccr4* KO. Adoptive transfer is followed by daily intranasal administration of HDM. After 5 days, the medLN is harvested and spatially sequenced with VisiumHD, which can then be used to train SpaceTravLR.

B. Schematic of Th2 lung-lymph migration patterns. Dendritic cells (DCs) in the lung activated upon encountering HDM migrate to the lymph, where they in turn activate the antigen-specific CD4 T cells through the *IL2*-signaling axis. Activated cells can differentiate into Th2 cells and migrate to the lung and cause inflammation.

C. Spatial organization of adoptively transferred cells and the anatomical regions (T cell zone, TB- order, B-cell follicles) in the publicly available Slide-seqV2 mouse lymph node harvested 3 days after adoptive transfer.

D. Ligand-receptor interactions with the highest SpaceTravLR coefficients. (interaction pairs with the C3 gene are not shown in this figure to focus on non-complement based immune interactions).

E. Th2 cell coefficients for the *Ccl5-Ccr4* interaction display the highest importance in the T cell zone which diminishes toward the B-cell follicles.

F. In silico *Ccr4* KO specifically within the Th2 cell compartment precipitates a marked depletion of Th2 populations exhibiting T-cell zone and T-B border transcriptomic identities. Conversely, this perturbation drives a substantial phenotypic shift, characterized by an expansion of Th2 cells adopting a B-cell follicle-associated transcriptomic program.

G. Novel intra-lymph migration patterns of Th2 cells as uncovered by SpaceTravLR. 1DER Th2 cells typically migrate to the T-cell zone of the lymph node, where the majority of the CD8 T cells reside. *Ccr4*-KO Th2 cells fail to migrate to these regions.

H. Control and **I.** Experimental *Ccr4*-KO 1DER VisiumHD samples harvested 4 days after adoptive transfer.

J. SpaceTravLR simulated *Ccr4* KO highlighting the Th2 cell locations (left) vs experimentally observed locations (right). Simulated Th2 cells are shown in red and experimentally observed Th2 cells are shown in orange.

K. Quantitative assessment of Th2 cell localization demonstrates a significant phenotypic shift upon *Ccr4* KO. Both knockout samples show a statistically higher occupancy of B-cell follicle niches and reduced infiltration of T-cell zone compartments (left), indicating a robust spatial preference shift away from the T-cell zone and T-B border environments (right).

L. Schematic illustrating the differential expression analysis workflow for cells adjacent to *Ccr4*-deficient Th2 cells. We identified consistent DEGs between control and both knockout samples and compared them with in-silico *Ccr4* perturbation predictions. Spatial neighborhoods were defined by the 15-nearest-neighbor radius, and target populations neighboring a perturbed Th2 cell were compared against a spatial baseline of the same cell type in non-perturbed regions.

M-O. Scatter plots quantifying alignment of the log2fold change in neighboring cells between simulated Th2-restricted *Ccr4* KO and experimental adoptive transfer KO. Data points are color-coded by prediction accuracy: green denotes concordant upregulation, red denotes concordant downregulation, and blue denotes discordant predictions. The simulations demonstrate high concordance with experimental results, yielding Spearman correlations of 0.23 in CD8 T cells (**M**), 0.92 in B cells (**N**), and 0.58 in DCs (**O**).

Extended Figure 1: SpaceTravLR's flexible framework can be applied across diverse perturbation settings and spatial contexts.

A. In this paper, SpaceTravLR has been applied to 16 different spatial transcriptomics datasets, 12 mouse and 4 human, and 8 different modalities including Slidetags, Xenium, Xenium 5K, SlideseqV2, XYZeqV2, Stereoseq, Merfish, and Atera. Accompanying the release of the paper, we will be releasing 2 novel datasets, including 2 XYZeqV2 murine kidneys, 1 VisiumHD lymph node, and 2 adoptive transfer *Ccr4*-KO lymph nodes.

Extended Figure 2: Rust implementation of SpaceTravLR is streamlined and accessible

A. The Rust refactoring of SpaceTravLR drastically improves ease of use by eliminating requirements for complex environmental dependency management and specialized programming knowledge. The integrated SpaceTravLR UI provides real-time monitoring of training progress, including status updates and estimated time to completion

B. SpaceTravLR was ported from Python/NumPy to Rust because the inner loops (Gaussian kernel, splash multiply, FISTA proximal steps) dominate runtime.

We demonstrate that our Rust implementation has much lower running time at every step of the pipeline, from training the CNN, computing the derivatives, and propagating the perturbation.

C. Comparison of the preprocessing time for a toy dataset consisting of 5000 cells and 4000 genes between the Scanpy (python) and Rust versions. The Rust version is much faster, at all steps of the preprocessing including QC filter, normalization, log-transformation, dimensionality reduction, clustering, and gene imputation.

Extended Figure 3: SpaceTravLR outperforms other perturbation methods in modeling cell-intrinsic perturbations.

A. Vector fields overlaid on the UMAP layouts, subset to CD4 T cells, showing the direction and magnitude of transcriptomic changes and its cosine similarity with pseudotime trajectory of CD4 T cell differentiation routes: from Th0 to Th1, Th0 to Tfh, and Th0 to Th2. These results demonstrate single-factor or combinatorial perturbations predicted by CellPLM, TahoeX1, SpaceFlow, CellOracle, and a Spatial Baseline, which is a small model created to predict the gene expression of target cells from neighboring cells (Methods).

B. CD4 T cell fate shifts in response to predictions of CellPLM, TahoeX1, SpaceFlow, SpaceTravLR, CellOracle, and a Spatial Baseline for *Il21* KO, *Il21* OE, *Bcl6* KO, *Bcl6* OE, *Prdm1* KO, *Prdm1* OE, and the combinatorial perturbations between *Il21* and *Bcl6*, as well as *Il21* and *Prdm1*. While CellOracle cannot model *Il21* or combinatorial perturbations, it can simulate its two downstream transcription factors individually. Consequently, these single-factor CellOracle predictions are displayed alongside the combinatorial subplots for baseline comparison. Arrows under each CD4 T cell subtype indicate the expected directional shift in response to the perturbation.

G. Quantification of the total number of directional cellular fate shifts induced across the eight previously described perturbation paradigms. Dashed bars denote the absolute number of correctly predicted directional shifts, while solid bars represent the subset of these shifts that achieved statistical significance.

Extended Figure 4: SpaceTravLR enables identification of novel targets with a limited gene panel and cell number

A-B. Spatial layout of the day 8 mouse small intestine profiled with Xenium colored by cell type (**A**) and crypt villi distance (**B**). Only a small fraction of the sample was trained (bottom)³.

C. *Tgfb1-Tgfb2* modulation of *Itgae* is correlated with the P14 CD8 T cell position along the crypt villi axis. The CNN scales the cell-type level SGL coefficients values depending on their spatial location. Here, the CNN scales down the *Tgfb1-Tgfb2* modulation for P14 CD8 T cells that are in the lower parts of the crypt villi axis.

D. Prioritization of LR interactions driving Trm commitment. Ranking of signaling pathways contributing to tissue-resident memory Trm cell differentiation, evaluated across a temporal trajectory (days 6, 8, 30, and 90). The prioritized interactions encompass established pathways known to govern Trm commitment and maturation.

E. P14 CD8 T cells were electroporated with Ccr5-targeting sgRNAs or non-targeting controls and adoptively transferred into mice prior to LCMV infection. Splenic and intestinal lymphocytes were harvested 7 days post-infection for phenotype and distribution analysis.

F. D7 intraepithelial lymphocytes in Cd19 (control) versus Ccr5 KO models, focusing on markers of tissue-resident memory Trm cell differentiation. The y-axes represent the frequency (%) of specific cell subsets defined by the expression of the retention marker CD103 and the early activation marker CD69. Ccr5 depletion significantly increases the frequency of the CD69+CD103- subset, indicating that Ccr5 signaling plays a measurable role in delaying Trm formation.

G-H. Spatial layout of the day 8 mouse small intestine profiled with VisiumHD used to train SpaceTravLR colored by cell type (**G**) and crypt villi distance (**H**). To ensure the integrity of the neighborhood ligand expression computation, the central regions were excluded from model training due to artifacts in the longitudinal unrolling of these sections.

I. Top LR pairs identified from SpaceTravLR trained on the VisiumHD sample with a larger gene panel. LR pairs also possible from the Xenium gene panel are colored in pink. The top LR interactions are all broad, and well characterized activation markers, most of which include genes not profiled in the original Xenium panel. Ccl5-Ccr5 interaction does not appear.

J-H. Ccr5 (**J**) and Ccl5 (**H**) expression in the VisiumHD sample is highly correlated with the expression of receptors ranked highly in the screening from I.

K. SpaceTravLR defines ligand modulator terms based on localized neighborhood expression. The spatial neighborhood expression of Ccl5 demonstrates a strong correlation with the neighborhood expression profiles of the top-ranked L-R pathways identified in (i).

Extended Figure 5. SpaceTravLR resolves subtle impacts of pharmacological perturbations on intercellular signaling networks

A. UMAP projection of cell types in a Xenium 5K mouse skin sample.

B-E. Spatial layouts (top) and top LR pairs promoting mast cell activation identified by SpaceTravLR in (**B**) a homeostatic (untreated) sample, (**C**) a sample treated with both Compound 48/80 and capsaicin, (**D**) a sample treated with Compound 48/80 alone, and (**E**) a sample treated with capsaicin alone. Bar colors indicate ligand contributions from distinct cell types in each sample.

F. Shapley scores for each component contributing to SpaceTravLR's estimation of target gene expression. TF, ligand-receptor, and CNN contribute substantial amounts of information, 36.8%, 39.4% and 22.3%, respectively.

G. Differential LR signaling analysis across the treatment conditions. Changes in LR pair rankings highlight SpaceTravLR's ability to capture condition-specific shifts in cell-cell communication following drug perturbation.

Extended Figure 6: SpaceTravLR captures phenotypic effects of global gene knockout in mouse embryo

A. Spatial map of E9.5 WT Slide-seqV2 data and different cell types used for training.

B. Schematic showing the expected somitic and neural developmental phenotype in WT embryo and its alterations after *Tbx6* KO.

C. UMAP of cell types in during normal development.

D. Ectopic tube cells from the real *Tbx6* knockout sample are projected into the same UMAP space as the wildtype cells. Vectors demonstrate SpaceTravLR predicted cell-state transitions resulting from *Tbx6* knockout.

E. SpaceTravLR-predicted rankings of DEGs following *in silico* *Tbx6* KO exhibit a strong correlation with the top 1,000 ground-truth experimental DEGs. This alignment outperforms both a permuted modulator control and CellOracle, an alternative zero-shot perturbation method but remains non-spatial and restricted to cell-intrinsic dynamics.

Extended Figure 7: Spatial microniche and modulation analysis in kidney tissue.

A. Dot plot of marker gene expression comparing regular and hypertonic macrophage populations, using an epithelial cell subset as a reference.

B. Top 10 strongly enriched gene programs identified by Nichecompass, displaying predicted areas of gene program activity (left) alongside the log-normalized expression of the most important constituent genes (right).

C. Top TF (top), LR (middle), and TF-ligand (bottom) modulator interactions exhibiting the highest Moran's I across the tissue. Distinct spatially varying regions of intrinsic and extrinsic signaling are visible, allowing for direct visual interpretation of localized modulation and regional gene importance.

D. Spatial layout of microniches identified by SpaceTravLR trained natively on the tissue (top left), BANKSY (bottom left), Leiden clustering on the COVET embedding space (top right), and NicheCompass (bottom right).

Extended Figure 8: Additional experimental perturb-seq data analysis and comparisons.

A. The number of cells of each cell type that received each target guide in the murine hippocampus perturb-seq experiment varies greatly. Many cell types did not receive enough of a certain gene guide to calculate the significant differentially expressed genes in their neighborhood.

B. Number of significant negative/downregulated (blue) and positive/upregulated (red) differentially expressed genes in each combination of gene guide, target cell (x-axis), and cell population (y-axis). Combinations highlighted in purple have either at least 5 upregulated and 5 downregulated DEGs or have at least 20 total DEGs. Combinations highlighted in green pass both thresholds and are most reliable to compute correlations between experimental and predicted log2fold change.

C. Tissue sections showing guide localization across multiple sectioning angles and orientations. Large colored regions indicate the final guide positions, while the heatmap overlay visualizes the spatial distribution of guide localization events. The number of neighbors around the cells receiving perturbations vary greatly. Together, these sections illustrate the model's ability to generalize across diverse tissue orientations and provide an intuitive view of localization density throughout the tissue.

D. Experimental vs SpaceTravLR log2fold change of the genes in the local neighborhood of the perturbed cell. All significant DEGs in the experimental data are shown, regardless of SpaceTravLR's predictions.

E. Experimental vs SpaceTravLR log2fold change of the genes in the local neighborhood of the perturbed cell in the MERFISH murine liver dataset. All genes that were predicted to be significant in the experimental data are shown, regardless of whether SpaceTravLR's predictions.

Extended Figure 9: SpaceTravLR identified functional microniches align with experimental spatial perturb-seq niches.

A. MC38 and Lung tumor H&E samples overlaid with (i) SpaceTravLR predicted microniches, (ii) Observed guide enrichment from the spatial perturbation dataset and (iii) SpaceTravLR predicted enrichment of guides in different niches

B. Correlation between experimentally observed odds ratio and SpaceTravLR predicted enrichment

C. Observed versus predicted shift in functional pathways following different perturbations across all tissue sections for MC38 and lung metastasis

D. SpaceTravLR versus BANKSY and negative controls, comparing the prediction performance for perturbation-based niche concordance, with SpaceTravLR showing the highest performance.

Extended Figure 10: Scalability of SpaceTravLR across massive cellular datasets.

A. UMAP (left) and spatial layout (right) of publicly available Atera human cervical cancer dataset

B. SpaceTravLR captures continuous, spatially varying regulatory dynamics across an extensive dataset of 717,576 cells. In cancer-associated fibroblasts (CAFs), the predicted activities for transcription factors SP1 and IRF8, alongside the COL4A5–SDC4 ligand-receptor (L-R) interaction regulating LUM expression, demonstrate highly localized enrichment patterns and distinct concentration gradients across the tissue architecture (top panel). A similar spatial heterogeneity is captured within the CD8+ T cell compartment, driven by the transcription factor Tp63 and the L-R interactions Adgre5–Cd55 and Gax6–Axl regulating Gzmk expression.

C. Top LR interactions driving tumor proliferation identified by SpaceTravLR align closely with known biology.

D. UMAP (left) and spatial layout (right) of publicly available Atera human breast cancer dataset

E. SpaceTravLR captures continuous, spatially varying regulatory dynamics across an extensive dataset of 170,057 cells. In ER+ tumors, the predicted activities for transcription factors FOXA1 and TFAP2A, alongside the LAMA5–ITGA6 LR interaction regulating CDH1 expression, demonstrate highly localized enrichment patterns and distinct concentration gradients across the tissue architecture (top panel). The TF Max exhibits higher modulation of Ccdc125 bordering the ER+ tumors while the TF Bach1 and the LR interaction between Grn and Sort1 exhibits an inversely proportional pattern.

F. Top LR interactions driving Muc1 niches identified by SpaceTravLR align closely with known biology.

Extended Figure 11. Additional analyses of Ccr4-KO perturbation experiments and SpaceTravLR results.

A. SpaceTravLR identifies a spatial gradient of Ccl5-Ccr4 and Ccl22-Ccr4 interactions, with the strongest signaling observed in the T cell zones.

B. Top LR interactions promoting Th2 identity identified by SpaceTravLR, subset by spatial zone. While Ccl5-Ccr4 and Ccl22-Ccr4 interactions rank highly across all three zones, the Ccl5-Ccr4 interaction ranks highest in the T cell zone, and both interactions are lower in the B cell zone. This demonstrates SpaceTravLR's ability to capture differing cell-cell communication events depending on spatial context.

C. Experimental pipeline of control and 1DER Ccr4 knockout experiment used to generate the three murine lymph node VisiumHD samples. Mean Fluorescence Intensity (MFI) demonstrates successful knockout.

D-G. Spatial layouts of the mouse lymph node samples used to prioritize and validate the effects of Ccr4 KO on Th2 cells. Tissues are colored by cell type and include **(D)** a Slide-seqV2 mouse lymph node, **(E)** a Visium HD control, **(F)** a Visium HD 1DER Ccr4 KO, and **(G)** another Visium HD 1DER Ccr4 KO sample.

H. Relative proportions of cell types in each sample.

I. Relative proportions of cells categorized into the B cell zone, T cell zone, or T-B border.

J. Dot plot of marker gene expression used to validate the Cell2Location cell-type deconvolution, given that the Visium HD control lymph node exhibited different cell type proportions compared to the other three samples.

K. Expression of Ccl5 and Ccl22 across the four lymph nodes. Ccl5 and Ccl22 are visibly concentrated near the T cell zones.

References

1. Armingol, E., Officer, A., Harismendy, O. & Lewis, N. E. Deciphering cell-cell interactions and communication from gene expression. *Nat. Rev. Genet.* **22**, 71–88 (2021).
2. Rao, A., Barkley, D., França, G. S. & Yanai, I. Exploring tissue architecture using spatial transcriptomics. *Nature* **596**, 211–220 (2021).
3. Jinek, M. *et al.* A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science* **337**, 816–821 (2012).
4. Dixit, A. *et al.* Perturb-seq: Dissecting molecular circuits with scalable single-cell RNA profiling of pooled genetic screens. *Cell* **167**, 1853-1866.e17 (2016).
5. Datlinger, P. *et al.* Pooled CRISPR screening with single-cell transcriptome readout. *Nat. Methods* **14**, 297–301 (2017).
6. Binan, L. *et al.* Simultaneous CRISPR screening and spatial transcriptomics reveal intracellular, intercellular, and functional transcriptional circuits. *Cell* **188**, 2141-2158.e18 (2025).

- 1921 7. Chen, K. H., Boettiger, A. N., Moffitt, J. R., Wang, S. & Zhuang, X. RNA imaging.
1922 Spatially resolved, highly multiplexed RNA profiling in single cells. *Science* **348**,
1923 aaa6090 (2015).
- 1924 8. Shen, K. *et al.* Spatial perturb-seq: single-cell functional genomics within intact tissue
1925 architecture. *Nat. Commun.* **17**, (2026).
- 1926 9. Nevue, A. A. *et al.* Spatially resolved, multimodal in vivo Perturb-seq using antibody-
1927 based cell hashing. *bioRxiv* (2026) doi:10.64898/2026.05.25.727765.
- 1928 10. Zhang, H. *et al.* Uncovering spatially resolved functional genomics with CRISPR
1929 screen sequencing. *Cell* (2026) doi:10.1016/j.cell.2026.04.049.
- 1930 11. Technology. <https://surveygenomics.com/tech>.
- 1931 12. Oliveira, M. F. de *et al.* High-definition spatial transcriptomic profiling of immune cell
1932 populations in colorectal cancer. *Nat. Genet.* **57**, 1512–1523 (2025).
- 1933 13. Chen, A. *et al.* Spatiotemporal transcriptomic atlas of mouse organogenesis using
1934 DNA nanoball-patterned arrays. *Cell* **185**, 1777–1792.e21 (2022).
- 1935 14. Lecun, Y., Bottou, L., Bengio, Y. & Haffner, P. Gradient-based learning applied to
1936 document recognition. *Proc. IEEE* **86**, 2278–2324 (1998).
- 1937 15. Russell, A. J. C. *et al.* Slide-tags enables single-nucleus barcoding for multimodal
1938 spatial genomics. *Nature* **625**, 101–109 (2024).
- 1939 16. Stebegg, M. *et al.* Regulation of the germinal center response. *Front. Immunol.* **9**,
1940 2469 (2018).
- 1941 17. Massoni-Badosa, R. *et al.* An atlas of cells in the human tonsil. *Immunity* **57**, 379-
1942 399.e18 (2024).
- 1943 18. Young, C. & Brink, R. The unique biology of germinal center B cells. *Immunity* **54**,
1944 1652–1664 (2021).
- 1945 19. Wang, B., Wang, M., Ao, D. & Wei, X. CXCL13-CXCR5 axis: Regulation in
1946 inflammatory diseases and cancer. *Biochim. Biophys. Acta Rev. Cancer* **1877**,
1947 188799 (2022).
- 1948 20. Vogelzang, A. *et al.* A fundamental role for interleukin-21 in the generation of T
1949 follicular helper cells. *Immunity* **29**, 127–137 (2008).
- 1950 21. Yi, J. S., Cox, M. A. & Zajac, A. J. Interleukin-21: a multifunctional regulator of
1951 immunity to infections. *Microbes Infect.* **12**, 1111–1119 (2010).
- 1952 22. Wu, Y. *et al.* The biological effects of IL-21 signaling on B-cell-mediated responses
1953 in organ transplantation. *Front. Immunol.* **7**, 319 (2016).
- 1954 23. Quast, I. *et al.* Interleukin-21, acting beyond the immunological synapse,
1955 independently controls T follicular helper and germinal center B cells. *Immunity* **55**,
1956 1414–1430.e5 (2022).
- 1957 24. Nakamura, S. *et al.* Expression and responsiveness of human interleukin-18 receptor
1958 (IL-18R) on hematopoietic cell lines. *Leukemia* **14**, 1052–1059 (2000).
- 1959 25. Nakanishi, K. Unique action of interleukin-18 on T cells and other immune cells. *Front.*
1960 *Immunol.* **9**, 763 (2018).

1961 26. Linterman, M. A. *et al.* IL-21 acts directly on B cells to regulate Bcl-6 expression and
1962 germinal center responses. *J. Exp. Med.* **207**, 353–363 (2010).
1963 27. Cosgrove, J. *et al.* B cell zone reticular cell microenvironments shape CXCL13
1964 gradient formation. *Nat. Commun.* **11**, 3677 (2020).
1965 28. Moser, B. CXCR5, the defining marker for follicular B helper T (TFH) cells. *Front.*
1966 *Immunol.* **6**, 296 (2015).
1967 29. Moens, L. & Tangye, S. G. Cytokine-mediated regulation of plasma cell generation:
1968 IL-21 takes center stage. *Front. Immunol.* **5**, 65 (2014).
1969 30. Dinarello, C. A., Novick, D., Kim, S. & Kaplanski, G. Interleukin-18 and IL-18 binding
1970 protein. *Front. Immunol.* **4**, 289 (2013).
1971 31. Allen, C. D. C., Okada, T. & Cyster, J. G. Germinal-center organization and cellular
1972 dynamics. *Immunity* **27**, 190–202 (2007).
1973 32. Gandhi, S. *et al.* Tahoe-x1: Scaling perturbation-trained single-cell foundation models
1974 to 3 billion parameters. *bioRxiv* (2025) doi:10.1101/2025.10.23.683759.
1975 33. Wen, H. *et al.* CellPLM: Pre-training of cell language model beyond single cells.
1976 *bioRxiv* (2023) doi:10.1101/2023.10.03.560734.
1977 34. Ren, H., Walker, B. L., Cang, Z. & Nie, Q. Identifying multicellular spatiotemporal
1978 organization of cells with SpaceFlow. *Nat. Commun.* **13**, 1–14 (2022).
1979 35. Kamimoto, K. *et al.* Dissecting cell identity via network inference and in silico gene
1980 perturbation. *Nature* **614**, 742–751 (2023).
1981 36. Sun, E. D., Buendia, A., Brunet, A. & Zou, J. SpatialProp: tissue perturbation
1982 modeling with spatially resolved single-cell transcriptomics. *bioRxivorg* (2025)
1983 doi:10.64898/2025.11.30.691355.
1984 37. Nurieva, R. I. *et al.* Generation of T follicular helper cells is mediated by interleukin-
1985 21 but independent of T helper 1, 2, or 17 cell lineages. *Immunity* **29**, 138–149 (2008).
1986 38. Choi, J. & Crotty, S. Bcl6-mediated transcriptional regulation of follicular helper T cells
1987 (TFH). *Trends Immunol.* **42**, 336–349 (2021).
1988 39. Smith, M. A. *et al.* PRDM1/Blimp-1 controls effector cytokine production in human
1989 NK cells. *J. Immunol.* **185**, 6058–6067 (2010).
1990 40. Cimmino, L. *et al.* Blimp-1 attenuates Th1 differentiation by repression of ifng, tbx21,
1991 and bcl6 gene expression. *J. Immunol.* **181**, 2338–2347 (2008).
1992 41. He, K. *et al.* Blimp-1 is essential for allergen-induced asthma and Th2 cell
1993 development in the lung. *J. Exp. Med.* **217**, e20190742 (2020).
1994 42. Oestreich, K. J. *et al.* Bcl-6 directly represses the gene program of the glycolysis
1995 pathway. *Nat. Immunol.* **15**, 957–964 (2014).
1996 43. Sheikh, A. A. & Groom, J. R. Transcription tipping points for T follicular helper cell
1997 and T-helper 1 cell fate commitment. *Cell. Mol. Immunol.* **18**, 528–538 (2021).
1998 44. Xu, K. *et al.* How neural networks extrapolate: From feedforward to Graph Neural
1999 Networks. *arXiv [cs.LG]* (2020) doi:10.48550/arXiv.2009.11848.

2000 45. Crotty, S. T follicular helper cell biology: A decade of discovery and diseases.
2001 *Immunity* **50**, 1132–1148 (2019).

2002 46. Gavrilidis, G. I., Vasileiou, V., Orfanou, A., Ishaque, N. & Psomopoulos, F. A mini-
2003 review on perturbation modelling across single-cell omic modalities. *Comput. Struct.*
2004 *Biotechnol. J.* **23**, 1886–1896 (2024).

2005 47. Kennedy, D. E. *et al.* Novel specialized cell state and spatial compartments within
2006 the germinal center. *Nat. Immunol.* **21**, 660–670 (2020).

2007 48. Nakagawa, R. & Calado, D. P. Positive selection in the light zone of germinal centers.
2008 *Front. Immunol.* **12**, 661678 (2021).

2009 49. Dominguez-Sola, D. *et al.* The FOXO1 transcription factor instructs the germinal
2010 center dark zone program. *Immunity* **43**, 1064–1074 (2015).

2011 50. Inoue, T. *et al.* The transcription factor Foxo1 controls germinal center B cell
2012 proliferation in response to T cell help. *J. Exp. Med.* **214**, 1181–1198 (2017).

2013 51. Cyster, J. G. Germinal centers: Gaining strength from the dark side. *Immunity* **43**,
2014 1026–1028 (2015).

2015 52. Ståhl, P. L. *et al.* Visualization and analysis of gene expression in tissue sections by
2016 spatial transcriptomics. *Science* **353**, 78–82 (2016).

2017 53. 10x Genomics. *Human Lymph Node, Spatial Gene Expression Dataset (Space*
2018 *Ranger 1.1.0)*. 10x Genomics [https://www.10xgenomics.com/datasets/human-](https://www.10xgenomics.com/datasets/human-lymph-node-1-standard-1-1-0)
2019 *lymph-node-1-standard-1-1-0* (2020).

2020 54. Stone, E. L. *et al.* ICOS coreceptor signaling inactivates the transcription factor
2021 FOXO1 to promote tfh cell differentiation. *Immunity* **42**, 239–251 (2015).

2022 55. Hedrick, S. M., Hess Michelini, R., Doedens, A. L., Goldrath, A. W. & Stone, E. L.
2023 FOXO transcription factors throughout T cell biology. *Nat. Rev. Immunol.* **12**, 649–
2024 661 (2012).

2025 56. Kerdiles, Y. M. *et al.* Foxo1 links homing and survival of naive T cells by regulating L-
2026 selectin, CCR7 and interleukin 7 receptor. *Nat. Immunol.* **10**, 176–184 (2009).

2027 57. Reina-Campos, M. *et al.* Tissue-resident memory CD8 T cell diversity is
2028 spatiotemporally imprinted. *Nature* **639**, 483–492 (2025).

2029 58. Cepek, K. L. *et al.* Adhesion between epithelial cells and T lymphocytes mediated by
2030 E-cadherin and the alpha E beta 7 integrin. *Nature* **372**, 190–193 (1994).

2031 59. Zhang, N. & Bevan, M. J. Transforming growth factor- β signaling controls the
2032 formation and maintenance of gut-resident memory T cells by regulating migration
2033 and retention. *Immunity* **39**, 687–696 (2013).

2034 60. Svensson, M. *et al.* CCL25 mediates the localization of recently activated
2035 CD8 α beta(+) lymphocytes to the small-intestinal mucosa. *J. Clin. Invest.* **110**,
2036 1113–1121 (2002).

2037 61. Klonowski, K. D. *et al.* Dynamics of blood-borne CD8 memory T cell migration in vivo.
2038 *Immunity* **20**, 551–562 (2004).

62. Ray, S. J. *et al.* The collagen binding alpha1beta1 integrin VLA-1 regulates CD8 T cell-mediated immune protection against heterologous influenza infection. *Immunity* **20**, 167–179 (2004).
63. Chen, C.-S. *et al.* The genesis of cutaneous neurogenic inflammation is a multi-step process requiring DC aggregation. *J. Invest. Dermatol.* **145**, e41 (2025).
64. McNeil, B. D. *et al.* Identification of a mast-cell-specific receptor crucial for pseudo-allergic drug reactions. *Nature* **519**, 237–241 (2015).
65. Wershil, B. K., Tsai, M., Geissler, E. N., Zsebo, K. M. & Galli, S. J. The rat c-kit ligand, stem cell factor, induces c-kit receptor-dependent mouse mast cell activation in vivo. Evidence that signaling through the c-kit receptor can induce expression of cellular function. *J. Exp. Med.* **175**, 245–255 (1992).
66. Pastwińska, J., Walczak-Drzewiecka, A., Łukasiak, M., Ratajewski, M. & Dastyk, J. Hypoxia regulates human mast cell adhesion to fibronectin via the PI3K/AKT signaling pathway. *Cell Adh. Migr.* **14**, 106–117 (2020).
67. Lundberg, S. & Lee, S.-I. A unified approach to interpreting model predictions. *arXiv [cs.AI]* (2017) doi:10.48550/arXiv.1705.07874.
68. Sampath Kumar, A. *et al.* Spatiotemporal transcriptomic maps of whole mouse embryos at the onset of organogenesis. *Nat. Genet.* **55**, 1176–1185 (2023).
69. Takemoto, T. *et al.* Tbx6-dependent Sox2 regulation determines neural or mesodermal fate in axial stem cells. *Nature* **470**, 394–398 (2011).
70. Efremova, M., Vento-Tormo, M., Teichmann, S. A. & Vento-Tormo, R. CellPhoneDB: inferring cell–cell communication from combined expression of multi-subunit ligand–receptor complexes. *Nat. Protoc.* **15**, 1484–1506 (2020).
71. Jin, S. *et al.* Inference and analysis of cell-cell communication using CellChat. *Nat. Commun.* **12**, 1–20 (2021).
72. Cang, Z. *et al.* Screening cell-cell communication in spatial transcriptomics via collective optimal transport. *Nat. Methods* **20**, 218–228 (2023).
73. Birk, S. *et al.* Quantitative characterization of cell niches in spatially resolved omics data. *Nat. Genet.* **57**, 897–909 (2025).
74. Singhal, V. *et al.* BANKSY unifies cell typing and tissue domain segmentation for scalable spatial omics data analysis. *Nat. Genet.* **56**, 431–441 (2024).
75. Jantsch, J. *et al.* Cutaneous Na⁺ storage strengthens the antimicrobial barrier function of the skin and boosts macrophage-driven host defense. *Cell Metab.* **21**, 493–501 (2015).
76. Ip, W. K. E. & Medzhitov, R. Macrophages monitor tissue osmolarity and induce inflammatory response through NLRP3 and NLRC4 inflammasome activation. *Nat. Commun.* **6**, 6931 (2015).
77. Muela-Zarzuela, I. *et al.* NLRP1 inflammasome promotes senescence and senescence-associated secretory phenotype. *Inflamm. Res.* **73**, 1253–1266 (2024).

78. Sandstrom, A. *et al.* Functional degradation: A mechanism of NLRP1 inflammasome activation by diverse pathogen enzymes. *Science* **364**, eaau1330 (2019).
79. Chen, Z. *et al.* Evidence for a role of macrophage migration inhibitory factor in vascular disease. *Arterioscler. Thromb. Vasc. Biol.* **24**, 709–714 (2004).
80. Müller, S. *et al.* Salt-dependent chemotaxis of macrophages. *PLoS One* **8**, e73439 (2013).
81. Calandra, T. & Roger, T. Macrophage migration inhibitory factor: a regulator of innate immunity. *Nat. Rev. Immunol.* **3**, 791–800 (2003).
82. Leng, L. *et al.* MIF signal transduction initiated by binding to CD74. *J. Exp. Med.* **197**, 1467–1476 (2003).
83. Haviv, D. *et al.* The covariance environment defines cellular niches for spatial inference. *Nat. Biotechnol.* 1–12 (2024).
84. Stickels, R. R. *et al.* Highly sensitive spatial transcriptomics at near-cellular resolution with Slide-seqV2. *Nat. Biotechnol.* **39**, 313–319 (2021).
85. Saunders, R. A. *et al.* Perturb-Multimodal: A platform for pooled genetic screens with imaging and sequencing in intact mammalian tissue. *Cell* **188**, 4790–4809.e22 (2025).
86. Zhao, Y. *et al.* Stereo-seq V2: Spatial mapping of total RNA on FFPE sections with high resolution. *Cell* **188**, 6554–6571.e21 (2025).
87. Sharma, P. & Allison, J. P. The future of immune checkpoint therapy. *Science* **348**, 56–61 (2015).
88. Ribas, A. & Wolchok, J. D. Cancer immunotherapy using checkpoint blockade. *Science* **359**, 1350–1355 (2018).
89. Wherry, E. J. T cell exhaustion. *Nat. Immunol.* **12**, 492–499 (2011).
90. Joyce, J. A. & Fearon, D. T. T cell exclusion, immune privilege, and the tumor microenvironment. *Science* **348**, 74–80 (2015).
91. Baessler, A. & Vignali, D. A. A. T cell exhaustion. *Annu. Rev. Immunol.* **42**, 179–206 (2024).
92. Bakr, M. N., Takahashi, H. & Kikuchi, Y. Analysis of melanoma gene expression signatures at the single-cell level uncovers 45-gene signature related to prognosis. *Biomedicines* **10**, 1478 (2022).
93. Gattinoni, L. *et al.* CTLA-4 dysregulation of self/tumor-reactive CD8⁺ T-cell function is CD4⁺ T-cell dependent. *Blood* **108**, 3818–3823 (2006).
94. Mittermüller, D. *et al.* PD-1 knockout on cytotoxic primary murine CD8⁺ T cells improves their motility in retrovirus infected mice. *Front. Immunol.* **15**, 1338218 (2024).
95. Leach, D. R., Krummel, M. F. & Allison, J. P. Enhancement of antitumor immunity by CTLA-4 blockade. *Science* **271**, 1734–1736 (1996).
96. Iwai, Y. *et al.* Involvement of PD-L1 on tumor cells in the escape from host immune system and tumor immunotherapy by PD-L1 blockade. *Proc. Natl. Acad. Sci. U. S. A.* **99**, 12293–12297 (2002).

97. Jung, I.-Y. *et al.* BLIMP1 and NR4A3 transcription factors reciprocally regulate antitumor CAR T cell stemness and exhaustion. *Sci. Transl. Med.* **14**, eabn7336 (2022).
98. Liu, X. *et al.* Genome-wide analysis identifies NR4A1 as a key mediator of T cell dysfunction. *Nature* **567**, 525–529 (2019).
99. Chen, J. *et al.* NR4A transcription factors limit CAR T cell function in solid tumours. *Nature* **567**, 530–534 (2019).
100. Hu, C., Zhu, L., Fan, P., Bu, X. & Zuo, C. TYRP1 defines a proliferative melanoma cell subpopulation, driving malignant progression and therapy resistance via the GPNMB-Notch1-SOX10/MITF axis. *J. Transl. Med.* (2026) doi:10.1186/s12967-026-08436-z.
101. Grosso, J. F. *et al.* LAG-3 regulates CD8⁺ T cell accumulation and effector function in murine self- and tumor-tolerance systems. *J. Clin. Invest.* **117**, 3383–3392 (2007).
102. Sakaguchi, S. *et al.* MAZR and Runx factors synergistically repress ThPOK during CD8⁺ T cell lineage development. *J. Immunol.* **195**, 2879–2887 (2015).
103. Andersen, L. *et al.* The transcription factor MAZR/PATZ1 regulates the development of FOXP3⁺ regulatory T cells. *Cell Rep.* **29**, 4447–4459.e6 (2019).
104. Cillo, A. R. *et al.* Blockade of LAG-3 and PD-1 leads to co-expression of cytotoxic and exhaustion gene modules in CD8⁺ T cells to promote antitumor immunity. *Cell* **187**, 4373–4388.e15 (2024).
105. Andrews, L. P. *et al.* LAG-3 and PD-1 synergize on CD8⁺ T cells to drive T cell exhaustion and hinder autocrine IFN- γ -dependent anti-tumor immunity. *Cell* **187**, 4355–4372.e22 (2024).
106. Atera: High throughput. Whole transcriptome. Single cell sensitivity. *10x Genomics* <https://www.10xgenomics.com/platforms/atera>.
107. Preview data: Atera in situ gene expression, FFPE human cervical cancer. *10x Genomics* <https://www.10xgenomics.com/datasets/atera-wta-ffpe-human-cervical-cancer>.
108. Preview data: Atera in situ gene expression, FFPE human breast cancer. *10x Genomics* <https://www.10xgenomics.com/datasets/atera-wta-ffpe-human-breast-cancer>.
109. Zaha, D. C. *et al.* Luminal versus non-luminal breast cancer CDH1 immunohistochemical expression. *Rev. Chim.* **70**, 465–469 (2019).
110. Sato, H., Higashi, S. & Miyazaki, K. Amino-terminal fragments of laminin γ 2 chain stimulate migration of metastatic breast cancer cells by interacting with CD44. *Clin. Exp. Metastasis* **32**, 405–415 (2015).
111. Siroy, A. *et al.* MUC1 is expressed at high frequency in early-stage basal-like triple-negative breast cancer. *Hum. Pathol.* **44**, 2159–2166 (2013).
112. Kufe, D. W. MUC1-C oncoprotein as a target in breast cancer: activation of signaling pathways and therapeutic approaches. *Oncogene* **32**, 1073–1081 (2013).

113. Kim, B. G. *et al.* Laminin-332-rich tumor microenvironment for tumor invasion in the interface zone of breast cancer. *Am. J. Pathol.* **178**, 373–381 (2011).
114. Hammad, H. & Lambrecht, B. N. The basic immunology of asthma. *Cell* **184**, 1469–1485 (2021).
115. He, K. *et al.* Spatial microniches of IL-2 combine with IL-10 to drive lung migratory TH2 cells in response to inhaled allergen. *Nat. Immunol.* **25**, 2124–2139 (2024).
116. Coquet, J. M. *et al.* Interleukin-21-producing CD4(+) T cells promote type 2 immunity to house dust mites. *Immunity* **43**, 318–330 (2015).
117. Isaikina, P. *et al.* Structural basis of the activation of the CC chemokine receptor 5 by a chemokine agonist. *Sci. Adv.* **7**, eabg8685 (2021).
118. Zhang, X.-F. *et al.* The regulatory network of the chemokine CCL5 in colorectal cancer. *Ann. Med.* **55**, 2205168 (2023).
119. Imai, T. *et al.* The T cell-directed CC chemokine TARC is a highly specific biological ligand for CC chemokine receptor 4. *J. Biol. Chem.* **272**, 15036–15042 (1997).
120. Kawasaki, S. *et al.* Intervention of thymus and activation-regulated chemokine attenuates the development of allergic airway inflammation and hyperresponsiveness in mice. *J. Immunol.* **166**, 2055–2062 (2001).
121. Liu, J. *et al.* CytoSignal detects locations and dynamics of ligand–receptor signaling at cellular resolution from spatial transcriptomic data. *Nat. Genet.* (2026) doi:10.1038/s41588-026-02624-9.
122. Li, S., Wang, R., Liu, S. & Li, S. C. Finding spatially variable ligand-receptor interactions with functional support from downstream genes. *Nat. Commun.* **16**, 7784 (2025).
123. Browaeys, R., Saelens, W. & Saeys, Y. NicheNet: modeling intercellular communication by linking ligands to target genes. *Nat. Methods* **17**, 159–162 (2020).
124. Dai, Z. *et al.* Geographically convolutional neural network weighted regression: a method for modeling spatially non-stationary relationships based on a global spatial proximity grid. *Geogr. Inf. Syst.* **36**, 2248–2269 (2022).
125. Simon, N., Friedman, J., Hastie, T. & Tibshirani, R. A Sparse-Group Lasso. *J. Comput. Graph. Stat.* **22**, 231–245 (2013).
126. Xia, C.-R., Cao, Z.-J., Tu, X.-M. & Gao, G. Spatial-linked alignment tool (SLAT) for aligning heterogenous slices. *Nat. Commun.* **14**, 7236 (2023).
127. La Manno, G. *et al.* RNA velocity of single cells. *Nature* **560**, 494–498 (2018).